



Output feedback motion control system for observation class ROVs based on a high-gain state observer: Theoretical and experimental results

Daniel de A. Fernandes^{a,*}, Asgeir J. Sørensen^a, Kristin Y. Pettersen^b, Décio C. Donha^c

^a Centre for Autonomous Marine Operations and Systems (AMOS), Department of Marine Technology, Norwegian University of Science and Technology (NTNU), Otto Nielsens veg 10, NO-7491 Trondheim, Norway

^b Centre for Autonomous Marine Operations and Systems (AMOS), Department of Engineering Cybernetics, Norwegian University of Science and Technology (NTNU), O.S. Bragstads Plass 2D, NO-7491, Trondheim, Norway

^c Department of Mechanical Engineering, Polytechnic School of the University of São Paulo (Poli/USP), Av. Prof. Mello Moraes, 2231, 05508-900 São Paulo, Brazil

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ABSTRACT

This paper proposes a motion control system for observation class ROVs. It is essentially an output feedback control system composed of a MIMO PID controller, which is aided by reference feedforward, and a high-gain observer. Feedback linearisation of the plant dynamics is also performed. Four degrees-of-freedom are controlled (surge, sway, heave, and yaw). Stability and satisfactory performance are attained under suitable and smooth reference trajectories, despite the presence of unmodelled plant dynamics, plant parameter variations, measurement errors and noise, and environmental disturbances. Results from full-scale sea trials based on NTNU's ROV Minerva are presented and discussed.

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1. Introduction

Observation class remotely operated vehicles (ROVs) are used worldwide as carriers of imagery devices, e.g. cameras, echo sounders, hyperspectral imagers, and sonars, for industrial, military, and research activities, e.g. inspections, surveys, mapping, and monitoring. They can also perform light-duty intervention tasks. Some application examples can be found in Fillinger and Funke (2013), Marsh, Copley, Huvenne, Tyler, and the Isis ROV facility (2013), Amado-Filho et al. (2012), Christ and Wernli (2007), Ludvigsen, Sortland, Johnsen, and Singh (2007), Singh, Roman, Pizarro, Eustice, and Can (2007), and Yoerger, Bradley, Walden, Singh, and Bachmayer (1998). ROVs are typically fully actuated vehicles with open-frame structures, which are teleoperated from support vessels through umbilical cables transmitting power, commands, and data. Regardless of the application, accurate motion control is required for capturing usable images at the

proper pace to construct consistent representations of objects or environments of interest.

A motion control system (MCS) for an ROV is typically composed of navigation, guidance, and control systems. The navigation system determines the position, attitude, and velocities of the controlled ROV through direct measurement or reconstruction via state estimation. The guidance system generates suitable and sufficiently smooth reference trajectories. It also generates the corresponding reference attitude, velocities, and accelerations along the reference trajectories. The control system determines the propulsion and steering forces and moments the propulsion system is to deliver in order for the controlled ROV to track the references.

This paper proposes an MCS with dynamic positioning (DP) and trajectory tracking capabilities for observation class ROVs, see Fig. 1. It essentially consists of an output feedback control system composed of a high-gain state observer (HGSO) and a multiple-input-multiple-output (MIMO) proportional-integral-derivative (PID) controller aided by reference feedforward. Feedback linearisation of the plant dynamics is also performed by the MCS. The MCS is synthesised based on a control plant model (CPM), i.e. based on a simplified version of the process plant model (PPM) (Sørensen, 2013, 2005).

The present work is an extension of the partial results previously reported in Fernandes, Sørensen, and Donha (2013, 2014), where the

* Corresponding author. Tel.: +47 735 51 461.

E-mail addresses: dalmeidafernandes@gmail.com,

daniel.fernandes@ntnu.no (D.d.A. Fernandes),

asgeir.sorensen@ntnu.no (A.J. Sørensen),

kristin.y.pettersen@itk.ntnu.no (K.Y. Pettersen), decdonha@usp.br (D.C. Donha).

Nomenclature

List of acronyms and abbreviations

BF	body-fixed
CB	centre of buoyancy
CG	centre of gravity
CPM	control plant model
DoF(s)	degree(s)-of-freedom
DP	dynamic positioning
DVL	Doppler velocity log
EKF	extended Kalman filter
GES	globally exponentially stable
HGSO(s)	high-gain state observer(s)
IMU	inertial measurement unit

ITAE	integral of time multiplied by absolute error
KYP	Kalman–Yakubovich–Popov
LTI	linear time-invariant
LTV	linear time-varying
MCS	motion control system
MIMO	multiple-input–multiple-output
NED	north-east-down
PID	proportional-integral-derivative
PPM	process plant model
ROV(s)	remotely operated vehicle(s)
SISO	single-input–single-output
SO(N)	special orthogonal group of order $N \in \mathbb{N}$
SPR	strictly positive real
SS(N)	skew-symmetric group of order $N \in \mathbb{N}$

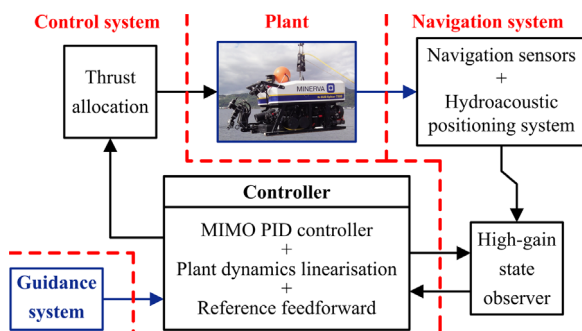


Fig. 1. Block diagram of the output feedback control system.

former introduces the MCS, and the latter reports details about its implementation in practice. The major enhancements with respect to both preceding works are the following:

- A more detailed CPM is provided and discussed.
- A tuning method for the PID controller is suggested.
- Further details on the HGSO are provided and discussed.
- A more thorough stability proof is provided. It only requires the mass and damping matrices of the CPM to be positive definite; they have to be neither diagonal nor symmetric.

The main contributions found in this paper are the following:

- An HGSO was tested in practice in an MCS for observation class ROVs with positive initial results.
- The controller tuning method was tested in practice. It is simple and useful, although attaining accurate motion control with smooth thruster action demands some tuning effort.
- A vector collecting current-generated perturbing forces and moment was derived based on the CPM. It provides useful information on the sea current. It can be used to help estimating the sea current, and to help determining the needed capacity of the propulsion system.

The classical PID control is often preferred for motion control of ROVs, and likewise for industrial applications in general (Smallwood & Whitcomb, 2004; Roberts, Sutton, Zirilli, & Tiano, 2003; Hsu, Costa, Lizarralde, & da Cunha, 2000), despite the availability of several other control techniques. Although the PID control has its limitations to cope with highly coupled multivariable nonlinear and time-varying dynamics, it is often preferred due to its synthesis and implementation simplicity, apart from being fully understood and having been

extensively and successfully tested in practice for almost one century since it was introduced (Åström & Hägglund, 2011; Franklin, Powell, & Emami-Naeini, 2009; Roberts et al., 2003; Friedland, 1986). The performance limitations can be sought to be circumvented by combining other techniques with the basic PID algorithm, whereupon satisfactory PID-based controllers can be obtained. Among the possibilities, the linearisation of the plant dynamics can often be attained through state feedback or reference feedforward, or the otherwise fixed controller gains can be varied online either discretely, through gain-scheduling, or continuously, through adaptive mechanisms, e.g. Lyapunov-based. Advanced control techniques are more capable of coping with the ROVs' dynamics, while also rejecting environmental disturbances. Still, accurate nominal models are usually needed in order for such techniques to yield the expected superior performance. Hence, a great effort may be necessary to accurately model and identify the hydrodynamic and actuator parameters, apart from also estimating the uncertainty boundaries of such parameters. A large number of examples of successful PID-based controllers can be found in literature, e.g. Sørensen (2013), Martin and Whitcomb (2012), Omerdic, Toal, Nolan, and Ahmad (2012, chap. 2), Sørensen, Dukan, Ludvigsen, Fernandes, and Candeloro (2012, chap. 6), Fossen (2011), Hoang and Kreuzer (2007), Smallwood and Whitcomb (2004), Caccia, Bono, Bruzzone, and Veruggio (2003), Hsu et al. (2000), Park, Chung, and Yuh (2000), Perrier and Canudas-de-Wit (1996), and references therein.

The vast majority of the marine output control systems are based on the benchmark extended Kalman filter (EKF), an extension to nonlinear systems of the optimal (minimum variance) linear-quadratic estimator Kalman filter, which has been extensively and successfully tested in practice for over half century since it was introduced (Sørensen, 2013; Fossen, 2011). Tuning an EKF may be a hard and time-consuming task due to its numerous covariance parameters, which may be difficult to relate to physical quantities (Fossen, 2011). Other nonlinear techniques, e.g. integrator backstepping and passivity, can be interesting alternatives for asymptotic state estimation (Fossen, 2011). An integrator-backstepping-based observer is easier to tune than an EKF. A passivity-based observer is easier to tune than both previous algorithms, requires a less detailed plant model to work satisfactorily, and allows design simplifications based on a version of the separation principle that holds for certain classes of nonlinear systems (Fossen, 2011; Khalil, 2002). An HGSO can robustly estimate the unmeasured states while it asymptotically rejects disturbances. It also permits the use of the separation principle for certain classes of nonlinear systems (Khalil, 2002; Atassi & Khalil, 2000; Esfandiari & Khalil, 1992). Its machine code needs less memory space to be stored, and also runs faster, than that of the EKF. It is found to be relatively easier to tune than the EKF tested earlier in Sørensen et al.

(2012). HGSOs currently represent an active research area, see Khalil and Praly (2014) and the references therein.

The paper is organised as follows. Section 2 introduces the CPM. Sections 3–5 concern the guidance, control, and navigation systems, respectively. Section 6 presents and discusses selected experimental results. Section 7 presents concluding remarks. Appendix A introduces the NTNU's ROV Minerva.

2. Control plant model

The proposed CPM concerns solely the controlled degrees-of-freedom (DoFs), namely surge, sway, heave, and yaw. It is based on the extensively used models found in Sørensen (2013), Fossen (2011), and Lewandowski (2004). The nomenclature defined by SNAME (1950) is adopted throughout. Particular attention is drawn to the hydrodynamic effects. The explicit dependence on time is omitted for the sake of simplicity.

The CPM is built upon these general premises: (i) the ROV is fully actuated regarding the four controlled DoFs; (ii) the two uncontrolled DoFs – roll and pitch – are self-stable by the design of the ROV (metacentric stability) and are not actuated; (iii) furthermore, both uncontrolled DoFs span negligibly narrow ranges, i.e. $|\phi| \leq 5^\circ$ (roll angle) and $|\theta| \leq 5^\circ$ (pitch angle), such that, in practice, the corresponding dynamics can be omitted from a (complete) dynamic model which would include six DoFs; (iv) the locations of the centre of gravity (CG) and of the centre of buoyancy (CB) are fixed; (v) the ROV operates deeply submerged, i.e. below the wave-affected zone; (vi) the velocity and the orientation of the sea current vary slowly enough to be taken as constant; and (vii) the fluid is irrotational, of constant and uniform density, and of infinite extent.

The CPM is defined as

$$\begin{cases} \dot{\eta} = J(\psi)\nu \\ \dot{M}\nu = -C(\nu)\nu - D_L\nu - D_Q|\nu|\nu + g + c + \tau \end{cases} \quad (1)$$

where $J(\psi) \in SO(4)$ (special orthogonal group of order 4) is a transformation matrix transforming the velocity vector $\nu = [u, v, w, r]^T$ given in the body-fixed (BF) coordinate frame into the velocity vector $\dot{\eta} = [\dot{n}, \dot{e}, \dot{d}, \dot{\psi}]^T$ given in the locally inertial north-east-down (NED) coordinate frame, see Fig. 2. The NED frame is usually used for locally flat Earth navigation of ROVs. The vector $|\nu|\nu := [|u|u, |v|v, |w|w, |r|r]^T \in \mathbb{R}^4$ represents signed quadratic velocities. The position and attitude (heading angle) vector $\eta = [n, e, d, \psi]^T$ gives the position from the chosen origin of the NED frame and the heading angle with respect to the N-axis of the NED frame. The inertia matrix $M \in \mathbb{R}^{4 \times 4} | M = M_{RB} + M_A > 0$ encompasses the mass matrix and the inertia tensor of the rigid-body ($M_{RB} \in \mathbb{R}^{4 \times 4} | M_{RB} > 0$), and also the hydrodynamic added mass matrix and its respective inertia tensor ($M_A \in \mathbb{R}^{4 \times 4} | M_A \geq 0$). The Coriolis-centripetal matrix $C(\nu) := C_{RB}(\nu) + C_A(\nu) \in SS(4)$ (skew-symmetric group of order 4) is directly derived from M . Naturally, everything affecting the entries of M also affect the entries of $C(\nu)$. The matrices

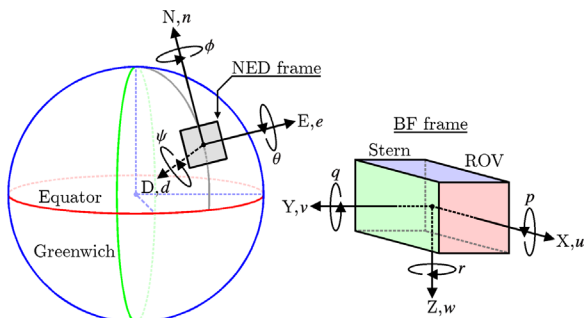


Fig. 2. NED coordinate frame and BF coordinate frame.

$D_L, D_Q \in \mathbb{R}^{4 \times 4} | D_L, D_Q > 0$ respectively collect linear and quadratic hydrodynamic damping coefficients. The vector of hydrostatic restoring forces $g = [0, 0, (W - B), 0]^T$ collects the weight force $W = mg$ acting upon the CG, and the buoyancy force $B = \rho \nabla g$ acting upon the CB, where m is the ROV's (dry) mass, g is the acceleration of gravity, ρ is the fluid density, and ∇ is the volume of fluid displaced by the ROV. The vector g is directly written in the BF frame under the premise that the roll and pitch dynamics are both neglectable, i.e. $|\phi| \approx |\theta| \approx 0^\circ \forall t \in [0, \infty)$ (Sørensen, 2013; Fossen, 2011). It is safer to have an ROV designed slightly positive buoyant, i.e. $B > W$, because it can then emerge to the surface in case a hard failure occurs in the MCS. The vector $c \in \mathbb{R}^4$ collects unmeasured (non-estimated either) current-generated perturbations. It is separately treated in Section 2.1. It is worthy noticing that (1) is based only on body-fixed velocities, without taking the sea current velocities into account, as the speed and the direction of the sea current are neither measured nor estimated in this work. The vector $\tau \in \mathbb{R}^4$ represents the propulsion and steering forces and moment delivered by the propulsion system. It is separately treated in Section 2.2.

Remark. The modelling of the hydrodynamic damping based on D_L and D_Q is assumed to provide a satisfactory approximation of the actual damping. It presents an intermediate level of complexity between decoupled models, e.g. Sørensen et al. (2012), Fossen (2011), Smallwood and Whitcomb (2004), and Caccia, Indiveri, and Veruggio (2000), and fully coupled, hence more accurate, models, e.g. Martin and Whitcomb (2012, 2008). The latter typically have as many time-varying matrices collecting nonlinear damping coefficients as DoFs. There are also other ways of modelling the nonlinear damping, e.g. Sørensen (2013), Fossen and Ross (2012, chap. 2), Fossen (2011), and Lewandowski (2004).

2.1. Current-generated perturbations

The vector of sea current velocities can be more naturally expressed in the NED frame as the vector $\nu_c^{\text{NED}} = \nu_c^{\text{NED}}(\alpha_H, \alpha_V)$, which is defined as

$$\nu_c^{\text{NED}} := V_c [c(\alpha_H)c(\alpha_V) \quad s(\alpha_H)c(\alpha_V) \quad s(\alpha_V) \quad 0]^T \quad (2)$$

where $V_c \in \mathbb{R}$ [m/s] is the current speed, $s(i) := \sin(i)$ and $c(i) := \cos(i)$, $i \in \{\alpha_H, \alpha_V\}$, $\alpha_H \in \mathbb{R} | \alpha_H \in (-\pi, \pi]$ [rad] is an angle contained in the horizontal NE-plane that starts from the N-axis, and $\alpha_V \in \mathbb{R} | \alpha_V \in [-\pi/2, \pi/2]$ [rad] is an angle contained in a vertical plane that starts from the NE-plane. The last entry is zero because the sea current is assumed to be irrotational in the NED frame. Then, the following relations arise:

$$\nu_c := \nu_c^{\text{BF}} = J^T(\psi) \nu_c^{\text{NED}} \quad \text{and} \quad \dot{\nu}_c = \frac{d}{dt} [J^T(\psi)] \nu_c^{\text{NED}} \quad (3)$$

Notice that $\dot{\nu}_c^{\text{NED}} = 0$, as the sea current is assumed constant.

The hydrodynamic effects can be modelled as (Sørensen, 2013; Fossen, 2011; Lewandowski, 2004)

$$\tau_h := -M_A \dot{\nu}_r - C_A(\nu_r) \nu_r - D_L \nu_r - D_Q |\nu_r| \nu_r \quad (4)$$

where $\nu_r := \nu - \nu_c = [u_r, v_r, w_r, r]^T$ is the velocity vector of the ROV relative to the current velocity vector $\nu_c = [u_c, v_c, w_c, r_c]^T$ in (3), where $r_c = 0$ rad/s, because the current is assumed to be irrotational. The vector of signed quadratic velocities is correspondingly defined as $|\nu_r| \nu_r := [|u_r|u_r, |v_r|v_r, |w_r|w_r, |r|r]^T \in \mathbb{R}^4$.

The 16 elements (hydrodynamic derivatives) of each matrix M_A , D_L , and D_Q can all be expected to be non-zero and distinct, as the ROV has an open-frame structure with asymmetries in the shapes and distribution of its internal parts and components, and, in addition, cabled and tubed interconnections among the parts and components randomly running inwards. In practice, every such matrix entry may slightly and slowly vary over time about its nominal value, mostly depending on operating and environmental conditions (Fossen, 2011;

Lewandowski, 2004; Caccia et al., 2000; Ishidera et al., 1986). Martin and Whitcomb (2008) provide an example of complete model parameter identification regarding all horizontal DoFs.

The vector of current-generated perturbations $\mathbf{c}$ stems from the terms in $\boldsymbol{\tau}_h$ in (4). Notice that $\mathbf{c} \neq \mathbf{0}$ if and only if $\boldsymbol{\nu}_c \neq \mathbf{0}$. Based on (4), it is possible to derive terms purely dependent on $\boldsymbol{\nu}$ and terms also involving $\boldsymbol{\nu}_c$ and $\dot{\boldsymbol{\nu}}_c$. The vectors $\dot{\boldsymbol{\nu}}_r$ and $|\boldsymbol{\nu}_r|\boldsymbol{\nu}_r$ are resolved into $\dot{\boldsymbol{\nu}} - \dot{\boldsymbol{\nu}}_c$ and $|\boldsymbol{\nu}|\boldsymbol{\nu} + (|\boldsymbol{\nu} - \boldsymbol{\nu}_c|(\boldsymbol{\nu} - \boldsymbol{\nu}_c) - |\boldsymbol{\nu}|\boldsymbol{\nu})$, respectively. Here, $|\boldsymbol{\nu} - \boldsymbol{\nu}_c|(\boldsymbol{\nu} - \boldsymbol{\nu}_c) \in \mathbb{R}^4$ indicates a multiplication performed componentwise (Hadamard, or Schur, product), e.g. $|u - u_c|(u - u_c)$, $|v - v_c|(v - v_c)$, and so forth, in the same fashion the vector $|\boldsymbol{\nu}_r|\boldsymbol{\nu}_r$ is defined. Then, it is straightforward to eliminate from (4) the terms which are only dependent on $\boldsymbol{\nu}$.

The expansion of the term $\mathbf{C}_A(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r$ yields

$$\mathbf{C}_A(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r := \mathbf{C}_A(\boldsymbol{\nu})\boldsymbol{\nu} + \mathbf{C}_{A_1}(\boldsymbol{\nu})\boldsymbol{\nu}_c + \mathbf{C}_{A_2}(\boldsymbol{\nu}_c)\boldsymbol{\nu}_c \quad (5)$$

where the matrices $\mathbf{C}_{A_i}(\cdot) \in \mathbb{R}^{4 \times 4}$, $i \in \{1, 2\}$, are defined as

$$\mathbf{C}_{A_1}(\boldsymbol{\nu}) := - \begin{bmatrix} \boldsymbol{\nu}^T \mathbf{T}_X \mathbf{M}_A \\ \boldsymbol{\nu}^T \mathbf{T}_Y \mathbf{M}_A \\ \mathbf{0}_{1 \times 4} \\ \boldsymbol{\nu}^T (\mathbf{T}_N \mathbf{M}_A + \mathbf{M}_A^T \mathbf{T}_N) \end{bmatrix} \quad (6)$$

$$\mathbf{C}_{A_2}(\boldsymbol{\nu}_c) := \begin{bmatrix} \mathbf{0}_{3 \times 4} \\ \boldsymbol{\nu}_c^T \mathbf{T}_N \mathbf{M}_A \end{bmatrix} \quad (7)$$

where $\mathbf{T}_X$, $\mathbf{T}_Y$, $\mathbf{T}_N \in \mathbb{R}^{4 \times 4}$ are constant sparse matrices, such that $(\mathbf{T}_X)_{4,2} = (\mathbf{T}_N)_{2,1} = -1$ and $(\mathbf{T}_Y)_{4,1} = (\mathbf{T}_N)_{1,2} = 1$ are the only non-zero elements. Notice that, in particular, $\mathbf{T}_N \in \mathcal{SS}(4)$. The matrices $\mathbf{0}_{i \times 4} \in \mathbb{R}^{i \times 4}$, $i \in \{1, 3\}$, are both null.

Finally, based on the expression of $\boldsymbol{\tau}_h$ in (4) and the results derived above, the vector of current-generated perturbing forces and moment $\mathbf{c} = \mathbf{c}(\boldsymbol{\nu}, \boldsymbol{\nu}_c, \dot{\boldsymbol{\nu}}_c)$ is defined as

$$\mathbf{c} := \mathbf{M}_A \dot{\boldsymbol{\nu}}_c - \mathbf{C}_{A_1}(\boldsymbol{\nu})\boldsymbol{\nu}_c - \mathbf{C}_{A_2}(\boldsymbol{\nu}_c)\boldsymbol{\nu}_c + \mathbf{D}_L \boldsymbol{\nu}_c + \mathbf{D}_Q(|\boldsymbol{\nu}|\boldsymbol{\nu} - |\boldsymbol{\nu} - \boldsymbol{\nu}_c|(\boldsymbol{\nu} - \boldsymbol{\nu}_c)) \quad (8)$$

The vector $\mathbf{c}$ provides useful information on the sea current. It can be used to estimate $\boldsymbol{\nu}_c$, but such a goal is not pursued in the present work. It can also be used in the design of the ROV to help determining the needed capacity of the propulsion system.

2.2. Propulsion system

Marine thrusters cannot be modelled by finite-dimensional models based on first principles (Sørensen, 2013). In particular, the complex nature of the propeller–fluid interaction makes it difficult to accurately model a thruster. Combinations of analytical and semi-empirical techniques are typically employed to produce simplified models. A static mapping relating the angular velocity of a propeller disc $n_p \in \mathbb{R}$ [rad/s] to the delivered thrust T [N] is a commonly made simplification (Sørensen, 2013; Caccia et al., 2000; Ishidera et al., 1986). In fact, the typical electromechanical time constants of the thrusters are negligible in comparison with the time constants of the overall dynamics of the marine crafts (Caccia et al., 2000).

It is then assumed that the thrust can be approximated by

$$T := k_T(n_p)|n_p|n_p \quad (9)$$

where $k_T(n_p) \in \mathbb{R}_{>0}$ is the thrust mapping, a nonlinear function of n_p that is determined in tank tests performed with the fully assembled ROV. It thus encompasses the propulsion losses.

The thrust configuration matrix of the propulsion system is given by $\mathbf{T}_p \in \mathbb{R}^{q \times p}$, $p \geq q$, where q is the dimension of the configuration space (the number of actuated DoFs), and p is the number of thrusters to actuate the configuration space.

The propulsion and steering vector is defined as

$$\boldsymbol{\tau} := \mathbf{T}_p [T_1 \ T_2 \ \dots \ T_p]^T \quad (10)$$

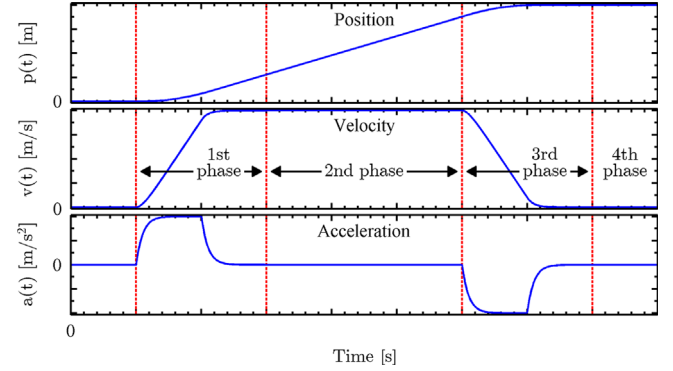


Fig. 3. Shapes of the references synthesised by the reference model.

where T_i , $i \in \mathbb{N} | i \in \{1, \dots, p\}$, is the thrust force delivered by the i th thruster of the ROV, which is modelled by (9).

3. Guidance system

The (open-loop) guidance system, see Fig. 1, generates suitable and sufficiently smooth reference trajectories parameterised by classes $\mathcal{C}^2$, $\mathcal{C}^1$, and $\mathcal{C}^0$, position, velocity, and acceleration references, respectively, which are synthesised by the reference model proposed by Fernandes, Dukan, and Sørensen (2012). The synthesised references are split into four consecutive phases, as depicted in Fig. 3, namely the acceleration phase (#1), the constant velocity phase (#2), the deceleration phase (#3), and the constant position phase (#4). Reference models are commonly employed in reference trajectory tracking control systems to improve the transient responses (Åström & Häggglund, 2011; Khalil, 2002). The guidance system is not detailed in the paper.

The full vector of references $[\boldsymbol{\eta}_R^T, \boldsymbol{\nu}_R^T, \dot{\boldsymbol{\nu}}_R^T]^T \in \mathbb{R}^{12}$ generated by the guidance system is composed of the position and heading angle reference vector $\boldsymbol{\eta}_R := [n_R, e_R, d_R, \psi_R]^T$ defined in the NED coordinate frame, of the velocity reference vector $\boldsymbol{\nu}_R := [u_R, v_R, w_R, r_R]^T$ defined in the BF coordinate frame, and of the acceleration reference vector $\dot{\boldsymbol{\nu}}_R(t) = (d/dt)[\boldsymbol{\nu}_R(t)]$, where $\dot{\boldsymbol{\eta}}_R(t) = (d/dt)[\boldsymbol{\eta}_R(t)] = \mathbf{J}(\psi(t))\boldsymbol{\nu}_R(t)$, and $t \in \mathbb{R}_{\geq 0}$ is the time.

4. Control system

4.1. Introduction

The control system performs two tasks: (i) linearisation of the plant dynamics; and (ii) asymptotic tracking of the reference vector $\mathbf{x}_R := [\boldsymbol{\eta}_R^T, \boldsymbol{\nu}_R^T]^T \in \mathbb{R}^8$. The first task is carried out aiming at helping the performance of the second task, and also at enabling the use of the linear HGSO, which is treated in Section 5.

4.2. Separation principle

It is assumed throughout Section 4 that the plant state vector $\mathbf{x} := [\boldsymbol{\eta}^T, \boldsymbol{\nu}^T]^T \in \mathbb{R}^8$ is available for feedback as if measured, whereas it is in reality the estimated vector $\hat{\mathbf{x}} := [\hat{\boldsymbol{\eta}}^T, \hat{\boldsymbol{\nu}}^T]^T \in \mathbb{R}^8$ supplied by the HGSO. This assumption is supported by a separation principle based on the use of the HGSO. Such a separation principle holds for certain classes of nonlinear systems (Khalil, 2002; Atassi & Khalil, 2000; Esfandiari & Khalil, 1992), including the presently proposed output feedback control system, given that: (i) the controlled plant is left-invertible and minimum-phase; and (ii) the state feedback control system is globally bounded in the state vector. The left-inversion problem characterises structural properties of a model with regard to estimation and reconstruction of its state vector (Estrada, García, Malabre, & García,

2007). Exact asymptotic tracking cannot be attained by non-minimum-phase systems (Franklin et al., 2009). Furthermore, such systems are frequently the source of considerable difficulty in synthesising well-behaved state feedback control systems (Friedland, 1986). The controlled plant is left-invertible and minimum-phase if its linearised version also has both characteristics (Esfandiari & Khalil, 1992). This is indeed the case in the present work, as it is shown later in this subsection. It is shown in Section 4.4 that the (state feedback) control system is globally exponentially stable (GES). This means that its region of attraction is the whole vector space $\mathcal{D} \equiv \mathbb{R}^{12}$, which is an equivalent condition to that of being globally bounded in the state vector. Finally, it is shown in Section 5.5 that the proposed output feedback control system is also GES.

Consider the following linear time-invariant (LTI) version of the CPM in (1) written in the state space

$$\begin{cases} \dot{\mathbf{z}} = \mathbf{A}\mathbf{z} + \mathbf{B}(\mathbf{c} + \boldsymbol{\tau}) \\ \mathbf{y} = \mathbf{C}\mathbf{z} \end{cases} \quad (11)$$

where $\mathbf{z} \in \mathbb{R}^8$ is the state vector, $\mathbf{y} \in \mathbb{R}^4$ is the output vector, $\mathbf{c}$ is the perturbation vector defined in (8), and $\boldsymbol{\tau}$ is the propulsion and steering vector defined in (10). Notice that $\mathbf{c}$ and $\boldsymbol{\tau}$ are two input vectors of the system (11). Let the matrices be defined as

$$\mathbf{A} := \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ \mathbf{0} & -\mathbf{M}^{-1}\mathbf{D}_L \end{bmatrix}, \quad \mathbf{B} := \begin{bmatrix} \mathbf{0} \\ \mathbf{M}^{-1} \end{bmatrix} \quad \text{and} \quad \mathbf{C} := [\mathbf{I} \quad \mathbf{0}] \quad (12)$$

where $\mathbf{0}, \mathbf{I} \in \mathbb{R}^{4 \times 4}$ are the null and identity matrices. Notice that only $\mathbf{M}$ and $\mathbf{D}_L$ are kept from (1). How this is achieved is detailed within the next subsection. The matrix pair $(\mathbf{A}, \mathbf{B})$ in (12) is controllable, as the controllability matrix $\mathbf{C}_c := [\mathbf{B} \quad \mathbf{A}\mathbf{B} \quad \mathbf{A}^2\mathbf{B} \quad \dots \quad \mathbf{A}^7\mathbf{B}]$ has full row rank, given that $\det([\mathbf{B} \quad \mathbf{A}\mathbf{B}]) = \det(\mathbf{M}^{-2}) > 0$, for instance. The matrix pair $(\mathbf{A}, \mathbf{C})$ in (12) is observable, since the observability matrix $\mathbf{O} := [\mathbf{C}^T \quad \mathbf{A}^T\mathbf{C}^T \quad (\mathbf{A}^2)^T\mathbf{C}^T \quad \dots \quad (\mathbf{A}^7)^T\mathbf{C}^T]^T$ has full column rank, provided that $\det([\mathbf{C}^T \quad \mathbf{A}^T\mathbf{C}^T]) = \det(\mathbf{I}) = 1$, for instance.

Then, the LTI system (11) is (i) a minimal realisation, as it has no transmission zeros in the whole complex plane, whereupon it is a minimum-phase system; and (ii) left-invertible, as the transfer function matrix $\mathbf{G}(s) := \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} \in \mathbb{R}^{4 \times 4}$ has full rank $\forall s \in \mathbb{C}$. Equivalently, $\dim(\ker(\mathbf{G}(s))) = 0 \forall s \in \mathbb{C}$.

The separation principle, as enunciated by Khalil (2002), Atassi and Khalil (2000), and Esfandiari and Khalil (1992), is therefore assumed to hold for the present nonlinear system, as the conditions on both the CPM and the state feedback controller are satisfied. No proof is repeated herein, and the interested reader is referred to the cited works for details and the proof. The perturbation theory (Khalil, 2002; Kokotović et al., 1999), which deals with dynamical systems containing different scales, whether space or time, is commonly employed therein to demonstrate that the state estimation by the HGSO converges very fast towards the actual plant states, for high enough gains of the HGSO. Thus, two time scales coexist in the output feedback control system, namely the slower state feedback dynamics and the faster state estimation dynamics, such that the system performance under output feedback is almost indistinguishable from that under state feedback.

4.3. Control vector

The control vector is defined as

$$\mathbf{u} := \mathbf{u}_{\text{LIN}} + \mathbf{u}_{\text{PID}} + \mathbf{u}_{\text{FF}} \quad (13)$$

where $\mathbf{u}_{\text{LIN}}$ tackles to cancel out the nonlinear terms in the dynamics of (1), $\mathbf{u}_{\text{PID}}$ implements full state feedback, and $\mathbf{u}_{\text{FF}}$ provides reference feedforward. The three terms in (13) are going to be detailed in the sequence.

The thrust allocation block, see Fig. 1, converts the control vector $\mathbf{u}$ into p individual signals μ_i (either voltages or currents) controlling

the thruster servos, where $\boldsymbol{\mu} \in \mathbb{R}^p | \boldsymbol{\mu} := [\mu_1, \mu_2, \dots, \mu_p]^T$. The signals then produce the vector $\boldsymbol{\tau}$ in (1) through (10). They are computed as $\mu_i := \theta_i \text{sgn}(T_i) \sqrt{|T_i|/k_{T_i}(n_{p_i})}$, $i \in \{1, \dots, p\}$, based on (9), where $\theta_i \in \mathbb{R}_{>0} | \theta_i := \mu_i/n_{p_i}$ are scaling factors (inverse thruster servo gains) which relate the signals μ_i to the corresponding propeller angular velocities n_{p_i} , and T_i are the thrust forces wanted the propellers to deliver, such that $[T_1, T_2, \dots, T_p]^T := \mathbf{T}_p^* \mathbf{u}$. The thrust allocation algorithm consists of the constant mapping $\mathbf{T}_p^* := \mathbf{W}_p^{-1} \mathbf{T}_p^T (\mathbf{T}_p \mathbf{W}_p^{-1} \mathbf{T}_p^T)^{-1}$, which basically consists of the Moore–Penrose pseudo-inverse of the thrust configuration matrix of the propulsion system $\mathbf{T}_p$. The (typically diagonal) matrix $\mathbf{W}_p \in \mathbb{R}^{p \times p} | \mathbf{W}_p = \mathbf{W}_p^T > 0$ weights the thruster usage, so that it is possible to reach an optimal thrust allocation through the minimisation of the quadratic cost function $J_T := \min_{\|\boldsymbol{\theta}_T\|} (\boldsymbol{\theta}_T^T \mathbf{W}_p \boldsymbol{\theta}_T)$ subject to $\boldsymbol{\tau} - \mathbf{T}_p \boldsymbol{\theta}_T = \mathbf{0}$, where $\boldsymbol{\theta}_T := [T_1, T_2, \dots, T_p]^T$, see (10). Details on such unconstrained least-squares optimisation problem can be found in Fossen (2011) and Fossen and Sagatun (1991).

4.3.1. Linearisation of the dynamics of the CPM

The nonlinear terms $\mathbf{C}(\nu)\nu$ and $\mathbf{D}_Q|\nu|/\nu$ in (1) can theoretically be cancelled out by state feedback, as both satisfy the matching condition, i.e. they enter the dynamics at the same point the control vector $\boldsymbol{\tau}$ does (Khalil, 2002), or, equivalently, they belong to the same range space of the input matrix $\mathbf{B}$ in (12). The vector $\mathbf{g}$ also satisfies the matching condition, whereupon it can also theoretically be eliminated. Cancelling them out corresponds to applying the method of computed torque, which is popular in the field of robotics (Franklin et al., 2009).

The linearisation term is defined as

$$\mathbf{u}_{\text{LIN}} := \hat{\mathbf{C}}(\nu)\nu + \hat{\mathbf{D}}_Q|\nu|/\nu - \bar{\mathbf{g}} \quad (14)$$

where

$$\hat{\mathbf{C}}(\nu) := \mathbf{W}_C \circ \bar{\mathbf{C}}(\nu) \quad \text{and} \quad \hat{\mathbf{D}}_Q := \mathbf{W}_Q \circ \bar{\mathbf{D}}_Q \quad (15)$$

where $\bar{\mathbf{C}}(\nu)$, $\bar{\mathbf{D}}_Q$, and $\bar{\mathbf{g}}$ are the nominal expressions of $\mathbf{C}(\nu)$, $\mathbf{D}_Q$, and $\mathbf{g}$ implemented in the MCS. The nominal matrices $\bar{\mathbf{C}}(\nu)$ and $\bar{\mathbf{D}}_Q$ may slightly differ from their actual counterparts, since $\bar{\mathbf{C}}(\nu)$ and $\bar{\mathbf{D}}_Q$ have fixed entries, whereas both actual matrices may vary slowly, as stated in Section 2.1. The nominal matrices often have diagonal structures, as it is difficult to obtain accurate estimates of their off-diagonal elements (Sørensen, 2013; Fossen, 2011; Lewandowski, 2004; Caccia et al., 2000). The vector $\bar{\mathbf{g}}$ may also slightly differ from the actual $\mathbf{g}$. The matrices $\mathbf{W}_C$, $\mathbf{W}_Q \in \mathbb{R}^{4 \times 4}$ weight, i.e. scale, the entries of $\bar{\mathbf{C}}(\nu)$ and $\bar{\mathbf{D}}_Q$, respectively, because, in practice, it is better to have some small residual nonlinear effects, than overcompensating for the nonlinear terms. The symbol ‘ $\circ$ ’ in (15) represents the entrywise Hadamard, or Schur, product.

The vector $\mathbf{u}_{\text{LIN}}$ represents the best endeavour towards linearising the dynamics of the CPM. However, the nonlinearities can never be exactly cancelled out, because the CPM always contains modelling simplifications and parameter uncertainties.

4.3.2. Asymptotic trajectory tracking

The control objective is to achieve $\|\mathbf{x}_R - \mathbf{x}\| \rightarrow 0$ as $t \rightarrow \infty$. The task is performed by a MIMO PID controller aided by reference feedforward (anticipative action). PID controllers fit well applications where sufficiently damped second-order dynamics are dealt with (Åström & Häggglund, 2011). Asymptotic tracking needs feedforward action to ensure that the needed control effort to achieve the goal is provided (Åström & Häggglund, 2011; Franklin et al., 2009; Friedland, 1986).

The feedback term is defined as

$$\mathbf{u}_{\text{PID}} := \mathbf{K}_P \mathbf{J}^T(\psi) \mathbf{e}_\eta + \mathbf{K}_I \mathbf{J}^T(\psi) \mathbf{e}_I + \mathbf{K}_D \mathbf{e}_v \quad (16)$$

where $\mathbf{e}_I := [\mathbf{e}_I^T, \mathbf{e}_\eta^T, \mathbf{e}_v^T]^T$ is the tracking error vector, whose components are the position tracking error vector $\mathbf{e}_\eta := \boldsymbol{\eta}_R - \boldsymbol{\eta}$, the velocity tracking

error vector $\mathbf{e}_\nu := \boldsymbol{\nu}_R - \boldsymbol{\nu}$, and the integral of $\mathbf{e}_\eta$ evaluated over time $\mathbf{e}_\eta := \int_0^t \mathbf{e}_\eta(\tau) d\tau$. The diagonal matrices $\mathbf{K}_P, \mathbf{K}_I, \mathbf{K}_D \in \mathbb{R}^{4 \times 4} | \mathbf{K}_P, \mathbf{K}_I, \mathbf{K}_D > 0$ keep the proportional, integral, and derivative sets of gains of the controller. A suitable method for tuning the PID controller is suggested and discussed in Section 4.5.

The reference feedforward term is defined as

$$\mathbf{u}_{FF} := \widehat{\mathbf{D}}_L \boldsymbol{\nu}_R + \widehat{\mathbf{M}} \dot{\boldsymbol{\nu}}_R \quad (17)$$

where

$$\widehat{\mathbf{D}}_L := \mathbf{W}_L \circ \overline{\mathbf{D}}_L \quad \text{and} \quad \widehat{\mathbf{M}} := \mathbf{W}_M \circ \overline{\mathbf{M}} \quad (18)$$

where $\overline{\mathbf{D}}_L$ and $\overline{\mathbf{M}}$ are the nominal expressions of $\mathbf{D}_L$ and $\mathbf{M}$ implemented in the MCS. The nominal matrices $\overline{\mathbf{D}}_L$ and $\overline{\mathbf{M}}$ may slightly differ from their actual counterparts, since $\overline{\mathbf{D}}_L$ and $\overline{\mathbf{M}}$ have fixed entries, whereas the actual matrices may vary slowly, as stated in Section 2.1. Both nominal matrices often have diagonal structures, as it is difficult to obtain accurate estimates of their off-diagonal elements (Sørensen, 2013; Fossen, 2011; Lewandowski, 2004; Caccia et al., 2000). The matrices $\mathbf{W}_L, \mathbf{W}_M \in \mathbb{R}^{4 \times 4}$ weight, i.e. scale, the entries of $\overline{\mathbf{D}}_L$ and $\overline{\mathbf{M}}$, respectively, because, in practice, it is advantageous, in terms of the overall tracking performance, to consider that the plant has less damping and inertia than it actually has, as this may result in smoother control actions.

4.4. Tracking error dynamics

By substituting (14), (16), and (17) into (13), and then by substituting (13) into (1) through (10), i.e. by utilising the relation $\boldsymbol{\tau} = \mathbf{u}$ introduced in Section 4.3, and lastly by performing a few algebraic operations to the finally derived expression, the tracking error dynamics is found to be given by

$$\dot{\mathbf{e}}_t = \mathbf{T}_t(\psi) \mathbf{A}_t \mathbf{T}_t^T(\psi) \mathbf{e}_t + \mathbf{B}_\delta^t \delta_t(\nu) + \mathbf{B}_R^t [\boldsymbol{\nu}_R^T, \dot{\boldsymbol{\nu}}_R^T]^T - \mathbf{B}_\delta^t \mathbf{c} \quad (19)$$

where $\mathbf{T}_t(\psi) := \text{blockdiag}(\mathbf{J}(\psi), \mathbf{J}(\psi), \mathbf{I}) \in \text{SO}(12)$, and

$$\mathbf{A}_t = \begin{bmatrix} \mathbf{0} & \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I} \\ -\mathbf{M}^{-1} \mathbf{K}_I & -\mathbf{M}^{-1} \mathbf{K}_P & -\mathbf{M}^{-1} (\mathbf{D}_L + \mathbf{K}_D) \end{bmatrix} \quad (20)$$

$$\mathbf{B}_\delta^t = [\mathbf{0} \ \mathbf{0} \ \mathbf{M}^{-T}]^T \quad (21)$$

$$\mathbf{B}_R^t = \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \\ \mathbf{M}^{-1} (\mathbf{D}_L - \widehat{\mathbf{D}}_L) & \mathbf{I} - \mathbf{M}^{-1} \widehat{\mathbf{M}} \end{bmatrix} \quad (22)$$

where $\mathbf{0}, \mathbf{I} \in \mathbb{R}^{4 \times 4}$ are the null and identity matrices, and $\delta_t(\nu)$ lumps together all the mismatches which stem from the attempt to linearise the dynamics of (1) through state feedback under the inevitable presence of modelling errors in (14), such that

$$\delta_t(\nu) := (\mathbf{C}(\nu) - \widehat{\mathbf{C}}(\nu))\nu + (\mathbf{D}_Q - \widehat{\mathbf{D}}_Q)|\nu| \quad (23)$$

The more exactly the parameters of the CPM are known, the closer to zero the mismatch vector $\delta_t(\nu)$ and the reference input matrix $\mathbf{B}_R^t$ are, with $(\mathbf{W}_C)_{ij} = (\mathbf{W}_L)_{ij} = (\mathbf{W}_M)_{ij} = (\mathbf{W}_Q)_{ij} = 1$, $i, j \in \{1, 2, 3, 4\}$, i.e. with all entries of these weighting matrices equal to one. The vector of references $[\boldsymbol{\nu}_R^T, \dot{\boldsymbol{\nu}}_R^T]^T$ remains bounded $\forall t \in [0, \infty)$, and vanishes eventually. Practically exact cancelling of the restoring forces is assumed, i.e. $\bar{\mathbf{g}} - \mathbf{g} \approx \mathbf{0}$.

Consider the global diffeomorphism $\boldsymbol{\xi}_t := \mathbf{T}_t^T(\psi) \mathbf{e}_t$ (Khalil, 2002), based on the defined transformation $\mathbf{T}_t(\psi)$. Then, (19) can be conveniently rewritten as

$$\dot{\boldsymbol{\xi}}_t = \boldsymbol{\Xi}_t(\dot{\psi}) \boldsymbol{\xi}_t + \mathbf{B}_\delta^t \delta_t(\nu) + \mathbf{B}_R^t [\boldsymbol{\nu}_R^T, \dot{\boldsymbol{\nu}}_R^T]^T - \mathbf{B}_\delta^t \mathbf{c} \quad (24)$$

where

$$\boldsymbol{\Xi}_t(\dot{\psi}) := \mathbf{A}_t - \mathbf{T}_t^T(\psi) \dot{\mathbf{T}}_t(\psi) = \mathbf{A}_t + \dot{\psi} \text{blockdiag}(\mathbf{T}_N, \mathbf{T}_N, \mathbf{0}) \quad (25)$$

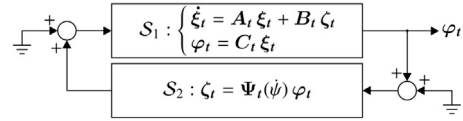


Fig. 4. Feedback connection between the auxiliary subsystems $\mathcal{S}_1$ and $\mathcal{S}_2$.

is a simpler linear time-varying (LTV) system matrix. The matrices $\mathbf{B}_R^t$ and $\mathbf{B}_\delta^t$ are not affected by the transformation $\mathbf{T}_t^T(\psi)$, so that $\mathbf{T}_t^T(\psi) \mathbf{B}_R^t = \mathbf{B}_R^t$ and $\mathbf{T}_t^T(\psi) \mathbf{B}_\delta^t = \mathbf{B}_\delta^t$ hold. The relation $\mathbf{S}_t(\dot{\psi}) := \dot{\psi} \text{blockdiag}(\mathbf{T}_N, \mathbf{T}_N, \mathbf{0}) = -\mathbf{T}_t^T(\psi) \dot{\mathbf{T}}_t(\psi)$, where $\dot{\psi} = r$ [rad/s] is the yaw rate and $\dot{\mathbf{T}}_t(\psi) := (d/dt)[\mathbf{T}_t(\psi)]$, is based on the matrix $\mathbf{T}_N$ defined in Section 2.1. The systems (19) and (24) are equivalent, although the latter shows clearer how heading angle changes affect the tracking error dynamics.

The stability of the origin $\boldsymbol{\xi}_t^* = \mathbf{0}$ (the only equilibrium point) of the unforced LTV system $\dot{\boldsymbol{\xi}}_t = \boldsymbol{\Xi}_t(\dot{\psi}) \boldsymbol{\xi}_t$ extracted from (24) is studied by suitably regarding the system as a feedback connection between the auxiliary subsystems $\mathcal{S}_1$ and $\mathcal{S}_2$, as seen in Fig. 4, where $\mathcal{S}_1$ is an LTI dynamical system and $\mathcal{S}_2$ is an LTV memoryless function (Khalil, 2002), and then applying the circle criterion (Khalil, 2002; Lozano et al., 2000). The origin of the unforced subsystem $\mathcal{S}_1$ is GES, once $\mathbf{A}_t$ is Hurwitz. The matrix $\mathbf{A}_t$ can always be rendered Hurwitz, since the pair $(\mathbf{A}, \mathbf{B})$ in (12) is controllable, as pointed out in Section 4.2.

Let the subsystem $\mathcal{S}_1$ be defined as indicated in Fig. 4. The matrix $\mathbf{B}_t := \mathbf{C}_t^T$, and $\mathbf{C}_t \in \mathbb{R}^{4 \times 12}$ is a sparse matrix whose only non-zero elements are $(\mathbf{C}_t)_{1,1} = (\mathbf{C}_t)_{2,2} = (\mathbf{C}_t)_{3,5} = (\mathbf{C}_t)_{4,6} = 1$. Both matrices have full rank, such that $\text{rank}(\mathbf{B}_t) = \text{rank}(\mathbf{C}_t) = 4$. The matrix pair $(\mathbf{A}_t, \mathbf{B}_t)$ is controllable, since the controllability matrix $\mathbf{C}_t := [\mathbf{B}_t \ \mathbf{A}_t \mathbf{B}_t \ \mathbf{A}_t^2 \mathbf{B}_t \ \dots \ \mathbf{A}_t^{11} \mathbf{B}_t]$ has full row rank, provided that $\det([\mathbf{B}_t \ \mathbf{A}_t^2 \mathbf{B}_t \ \mathbf{A}_t^4 \mathbf{B}_t]) \neq 0$, for instance. The matrix pair $(\mathbf{A}_t, \mathbf{C}_t)$ is observable, since the observability matrix $\mathcal{O}_t := [\mathbf{C}_t^T \ \mathbf{A}_t^T \mathbf{C}_t^T \ (\mathbf{A}_t^2)^T \mathbf{C}_t^T \ \dots \ (\mathbf{A}_t^{11})^T \mathbf{C}_t^T]^T$ has full column rank, given that $\det([\mathbf{C}_t^T \ (\mathbf{A}_t^2)^T \mathbf{C}_t^T \ (\mathbf{A}_t^4)^T \mathbf{C}_t^T]) \neq 0$, for instance. Recall that the hydrodynamic derivatives in $\mathbf{D}_L$ and $\mathbf{M}_A$, therefore those in $\mathbf{M}$, are all expected to be non-zero and distinct, as stated in Section 2.1. This fact guarantees that $\mathbf{C}_t$ and $\mathcal{O}_t$ have full rank, because $\mathbf{A}_t$ is composed of $\mathbf{D}_L$ and $\mathbf{M}^{-1}$. Then, the (Hurwitz) transfer function matrix $\mathbf{H}_t(s) := \mathbf{C}_t (s\mathbf{I} - \mathbf{A}_t)^{-1} \mathbf{B}_t \in \mathbb{R}^{4 \times 4}$ associated with $\mathcal{S}_1$, where $\mathbf{I} \in \mathbb{R}^{12 \times 12}$ is an identity matrix, is strictly positive real (SPR), as there exist for $\mathcal{S}_1$ a matrix $\mathbf{P}_t = \mathbf{P}_t^T > 0$, where $\mathbf{P}_t := \mathbf{I}$, a matrix $\mathbf{Q}_t \in \mathbb{R}^{12 \times 12} | \mathbf{Q}_t := \boldsymbol{\Gamma}_t^T \mathbf{Y}_t$, all going to be defined soon, and a constant $\varepsilon_t \in \mathbb{R} > 0 | \varepsilon_t > \max(|\lambda_i(\boldsymbol{\Lambda}_t)|)$, $i \in \{1, 2, \dots, 12\}$, which satisfy the relations seen in (26) regarding the Kalman–Yakubovich–Popov (KYP) lemma (Khalil, 2002; Lozano et al., 2000)

$$\begin{cases} \mathbf{A}_t^T \mathbf{P}_t + \mathbf{P}_t \mathbf{A}_t = -\mathbf{Q}_t - \varepsilon_t \mathbf{P}_t \\ \mathbf{P}_t \mathbf{B}_t = \mathbf{C}_t^T \end{cases} \quad (26)$$

where $\boldsymbol{\Gamma}_t \in \mathbb{R}^{12 \times 12}$ is an orthogonal matrix forming a basis of $\mathbb{R}^{12}$ that consists of eigenvectors of $(\mathbf{A}_t^T + \mathbf{A}_t)$, and $\boldsymbol{\Lambda}_t \in \mathbb{R}^{12 \times 12}$ is a diagonal matrix that is composed of the eigenvalues of the orthogonally diagonalisable symmetric matrix $(\mathbf{A}_t^T + \mathbf{A}_t)$ (Horn & Johnson, 2013), such that $\mathbf{A}_t^T + \mathbf{A}_t = \boldsymbol{\Gamma}_t \boldsymbol{\Lambda}_t \boldsymbol{\Gamma}_t^T$. The diagonal matrix $\mathbf{Y}_t \in \mathbb{R}^{12 \times 12} | \mathbf{Y}_t > 0$, where $(\mathbf{Y}_t)_{ii} = \sqrt{\varepsilon_t + (\boldsymbol{\Lambda}_t)_{ii}}$, $i \in \{1, 2, \dots, 12\}$. It can also be concluded that $\mathcal{S}_1$ is (state) strictly passive, once $\mathbf{H}_t(s)$ is SPR, according to Khalil (2002) and Lozano, Brogliato, Egeland, and Maschke (2000).

Let the matrix $\boldsymbol{\Psi}_t(\dot{\psi}) \in \mathbb{R}^{4 \times 4}$ of the subsystem $\mathcal{S}_2$ indicated in Fig. 4 be defined as

$$\boldsymbol{\Psi}_t(\dot{\psi}) := \dot{\psi} \text{blockdiag} \left(\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \right) \quad (27)$$

Then, $\boldsymbol{\varphi}_t^T \boldsymbol{\xi}_t = \boldsymbol{\varphi}_t^T \boldsymbol{\Psi}_t(\dot{\psi}) \boldsymbol{\varphi}_t = \boldsymbol{\xi}_t^T \mathbf{B}_t \boldsymbol{\Psi}_t(\dot{\psi}) \mathbf{C}_t \boldsymbol{\xi}_t = 0 \forall \boldsymbol{\psi} \in \mathbb{R}$, $\forall t \in [0, \infty)$, and $\forall \boldsymbol{\varphi}_t \in \mathbb{R}^4$ ($\boldsymbol{\xi}_t \in \mathbb{R}^{12}$), due to the skew-symmetry of $\boldsymbol{\Psi}_t(\dot{\psi})$. The relation $\mathbf{B}_t = \mathbf{C}_t^T$ is utilised above. Thus, it is immediate to conclude that $\boldsymbol{\varphi}_t^T \boldsymbol{\Psi}_t(\dot{\psi}) \boldsymbol{\varphi}_t$ lies in the sector $[0, \infty)$, and also that

$\mathcal{S}_2$ is a lossless passive subsystem, according to Khalil (2002) and Lozano et al. (2000).

It can thus be concluded, based on the application of the circle criterion, that ξ_t^* is GES (Th. 3.5 Lozano et al., 2000), once $\xi_t = \Xi_t(\psi)\xi_t$ is viewed as consisting of a feedback connection between the SPR transfer function matrix $H_t(s)$ (subsystem $\mathcal{S}_1$) and the lossless passive subsystem $\mathcal{S}_2$, as depicted in Fig. 4.

Finally, based on the concluded global stability of ξ_t^* , thereby implying that of $e_t^* = 0$, it can be inferred that the trajectory tracking control objective $\|x_R - x\| \rightarrow 0$ as $t \rightarrow \infty$ can always be attained. In particular, for persistent current-generated perturbations, i.e. $c \neq 0 \forall t \in [0, \infty)$, and a vanishing vector of references $[v_R^T, \dot{v}_R^T]^T$, under the inevitable presence of modelling errors, all solutions of (24), and those of (19) likewise, converge into a small ball $\mathbb{B}_t$ of radius $\rho_{\mathbb{B}_t} \in \mathbb{R}_{>0} | \rho_{\mathbb{B}_t} < 1$ ($\rho_{\mathbb{B}_t} \ll 1$ ideally) centred at $\xi_t^* \equiv e_t^*$. The radius $\rho_{\mathbb{B}_t}$ is directly proportional to $\|c\|$ and inversely proportional to the controller gains.

4.5. Tuning

Tuning a MIMO PID controller is a hard task in general, because it frequently involves a considerable number of variables. The complexity of the derived expressions often render the tuning problem difficult to analyse and solve analytically. Finding the optimal tuning is an even harder task, that may be practically unfeasible for higher-order systems. Numerical techniques, e.g. exhaustive search, genetic algorithms, and optimisation, among others, are thus commonly employed to find a satisfactory tuning. Common to all techniques is the fact that every candidate tuning has to be numerically evaluated, and the number of candidates grows exponentially with the size of the problem (combinatorial explosion), whereupon a prohibitively long time may be demanded to complete the tuning process.

It is thus suggested here to tune the MIMO PID controller based on the well-known optimal criterion named integral of time multiplied by absolute error (ITAE) that was originally developed for LTI single-input-single-output (SISO) systems (Åström & Hägglund, 2011; Franklin et al., 2009; Graham & Lathrop, 1953). This criterion is adapted and replicated to four decoupled SISO systems herein. The gains are defined as

$$\begin{cases} K_P := 4.94 \text{diag}(\widehat{M})\Omega^2 \\ K_I := \text{diag}(\widehat{M})\Omega^3 \\ K_D := 2.97 \text{diag}(\widehat{M})\Omega \end{cases} \quad (28)$$

where the numeric constants are those appearing in the denominator of the third-order ITAE prototype transfer function for zero tracking error of acceleration references (Graham & Lathrop, 1953). The matrix $\text{diag}(\widehat{M})$ keeps only the main diagonal of $\widehat{M}$ in (18), and the diagonal matrix $\Omega \in \mathbb{R}^{4 \times 4}$ is defined as

$$\Omega := 2\pi \left((\text{diag}(\widehat{M}))^{-1} W_\Omega \text{diag}(\widehat{D}_L) \right)^{-1} \quad (29)$$

where Ω is the multivariable version of the scalar counterpart $\omega = 2\pi/\tau$ [rad/s], where τ [s] is the natural motion period. The matrix $\text{diag}(\widehat{D}_L)$ keeps only the main diagonal of $\widehat{D}_L$ in (18), and the diagonal matrix $W_\Omega \in \mathbb{R}^{4 \times 4}$ weights the controller gains, so that the higher an entry of W_Ω , the lower the gains regarding the corresponding DoF. The use of the diagonal elements of $\widehat{M}$ and $\widehat{D}_L$ solely is strictly required in order to guarantee that $K_P, K_I, K_D > 0$.

This approach guarantees that all coefficients of the following 12th-order characteristic polynomial of A_t

$$p_{A_t}(s) = \det(I s^3 + M^{-1}((D_L + K_D)s^2 + K_P s + K_I)) \quad (30)$$

where $I \in \mathbb{R}^{4 \times 4}$ is an identity matrix, remain strictly positive, which is a necessary, although not sufficient, condition for stability of A_t

(Franklin et al., 2009; Friedland, 1986). The stability is assessed by finding the location of the eigenvalues of A_t , i.e. by finding the zeros of the characteristic equation $p_{A_t}(s) = 0$. This result holds as long as $M > 0$ and $D_L > 0$, independently of whether these matrices have or not diagonal structures. Both matrices M and D_L are defined within Section 2.

A thorough proof of the statement on the positiveness of the coefficients of $p_{A_t}(s)$ is omitted for the sake of brevity, although a hint for the simplest case can be suitably provided as follows: if all matrices in (30) are regarded as diagonal, then four independent third-order SISO systems arise, whereupon the expected result naturally arises by finding the zeros of $p_{A_t}(s) = 0$.

Robust stability against plant parameter variations can be attained at the expense of fairly intensive algebraic work on $p_{A_t}(s)$, which can be substantially speeded up with the aid of a computer, as the 12 controller gains ultimately depend only on the four entries of W_Ω . The intervals of $(W_\Omega)_{i,i}$, $i \in \{1, 2, 3, 4\}$, where A_t remains Hurwitz under presumed changes of the entries of M and D_L , with $M > 0$ and $D_L > 0$, can be conservatively found via the Routh–Hurwitz stability criterion (Franklin et al., 2009; Friedland, 1986) or, alternatively, via the Geršgorin's discs (Horn & Johnson, 2013). The latter is more restrictive, and has therefore limited usefulness in providing the desired intervals, despite its simplicity, and the lessened analysis complexity it demands.

Remark. The suggested approach provides an easier manner to tackle the tuning task, since the gains of the MIMO PID controller become ultimately dependent solely on the diagonal entries of W_Ω . This means that the gains can be readily calculated through the use of ((28) and (29)) once the elements of the main diagonal of both matrices M and D_L are identified in full-scale tests performed with the controlled ROV either in a tank or at the sea. The tuning problem thus resides in determining the appropriate entries of W_Ω , with the advantage offered by the adopted control architecture – MIMO PID controller, feedback linearisation of the plant dynamics, and reference feedforward working together – that each entry of W_Ω corresponds to a DoF which is independent from all other DoFs to a certain extent. As satisfactory performance must be sought, this objective underlies the process of deciding the values of the entries of W_Ω . However, prior to directly using any newly calculated set of gains K_P, K_I , and K_D , Eq. (30) has to be used to assess the stability of A_t in the manner described above, i.e. to determine the eigenvalues of A_t through finding the zeros of $p_{A_t}(s) = 0$, where the newly calculated set of gains has to yield stability in the worst-case scenario, given the presumed variation ranges of the entries of M and D_L . Finally, fine-tuning can be achieved by tweaking the gains after a satisfactory coarser tuning is achieved, always making sure that the gains yield stability.

5. Navigation system: high-gain state observer

5.1. Introduction

An HGSO is used in the MCS, see Fig. 1. This section then focuses on the application of the theory on HGSOs that is found in Khalil (2002), Atassi and Khalil (2000), and Esfandiari and Khalil (1992). Recalling its main features, an HGSO is a linear system based on the CPM that behaves approximately like a differentiator. It is robust to estimate the unmeasured states, while rejecting the effect of disturbances due to unmodelled dynamics and parameter uncertainties. It allows an output feedback control system to recover the performance achievable under state feedback when the gains are sufficiently high. It also produces initial impulse-like transient responses (peaking phenomenon), i.e. high-valued peaks of short duration which are proportional to the estimation errors. If the initial state of the controlled plant could be exactly known, then the initial estimation error would be zeroed, whereupon the initial peaks would be precluded. In practice, the

typical initial peaks can be at least drastically attenuated if the initial plant state can be accurately known.

5.2. Asymptotic state estimation

The estimation objective is to achieve $\|\mathbf{x} - \hat{\mathbf{x}}\| \rightarrow 0$ as $t \rightarrow \infty$, where $\hat{\mathbf{x}} := [\hat{\boldsymbol{\eta}}^T, \hat{\boldsymbol{\nu}}^T]^T \in \mathbb{R}^8$ is the estimated state vector. The HGSO is implemented here based on the CPM in (1), yet keeping only the linear part of the dynamics of (1), as it follows

$$\dot{\hat{\mathbf{x}}} = \mathbf{T}_e(\psi) \mathbf{A}_e \mathbf{T}_e^T(\psi) \hat{\mathbf{x}} + \mathbf{B}_e \mathbf{v} + \mathbf{T}_e(\psi) \mathbf{L} \mathbf{T}_e^T(\psi) (\mathbf{y} - \hat{\mathbf{y}}) \quad (31)$$

where $\mathbf{T}_e(\psi) := \text{blockdiag}(\mathbf{J}(\psi), \mathbf{I}) \in \text{SO}(8)$, and the remaining matrices are given by

$$\mathbf{A}_e := \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ \mathbf{0} & -\widehat{\mathbf{M}}^{-1} \widehat{\mathbf{D}}_L \end{bmatrix} \quad \text{and} \quad \mathbf{B}_e := \begin{bmatrix} \mathbf{0} \\ \widehat{\mathbf{M}}^{-1} \end{bmatrix} \quad (32)$$

$$\mathbf{L} := \begin{bmatrix} \epsilon^{-1} \mathbf{L}_{11} & \gamma \epsilon^{-1} \mathbf{L}_{12} \\ \epsilon^{-2} \mathbf{L}_{21} & \gamma \epsilon^{-2} \mathbf{L}_{22} \end{bmatrix} \quad (33)$$

where $\mathbf{0}, \mathbf{I} \in \mathbb{R}^{4 \times 4}$ are the null and identity matrices, $\mathbf{L}_{ij} \in \mathbb{R}^{4 \times 4}$, $i, j \in \{1, 2\}$, are tuning submatrices which compose the output injection matrix $\mathbf{L}$, the small constant $\epsilon \in \mathbb{R}_{>0} | \epsilon \ll 1$ is a global tuning parameter yielding the high-gains, $\gamma \in \mathbb{N} | \gamma \in \{0, 1\}$ indicates the availability of the velocity measurements, where $\gamma = 1$ means ‘available’, $\mathbf{v} \in \mathbb{R}^4 | \mathbf{v} := \mathbf{u}_{\text{PID}} + \mathbf{u}_{\text{FF}}$ is the control vector dedicated to the HGSO, see Fig. 1, $\mathbf{y} := [\boldsymbol{\eta}^T, \boldsymbol{\nu}^T]^T \in \mathbb{R}^8$ is the measurement vector, and $\hat{\mathbf{y}} := \mathbf{C}_e \hat{\mathbf{x}}$, where $\mathbf{C}_e \in \mathbb{R}^{8 \times 8}$ such that $\mathbf{C}_e := \text{blockdiag}(\mathbf{I}, \mathbf{I})$, is the estimated state vector actually feeding the control system in Section 4. It is worthwhile to realise the resemblance kept between the matrix pair formed by $\mathbf{A}_e$ and $\mathbf{B}_e$ in (32), and the corresponding matrix pair in (12).

5.3. Estimation error dynamics

By substituting (14), (16), and (17) into (13), and then by substituting (13) into (1) through (10), i.e. by utilising the relation $\boldsymbol{\tau} = \mathbf{u}$ introduced in Section 4.3, the closed-loop trajectory tracking control subsystem – 1st subsystem – is set. To derive the closed-loop state observation subsystem – 2nd subsystem, (16) and (17) are first added to yield $\mathbf{v}$, and then the 2nd subsystem is set by substituting $\mathbf{v}$ into (31). By subtracting the expression of the 2nd subsystem from the expression of the 1st subsystem, and lastly by performing a few algebraic operations, the estimation error dynamics is found to be given by

$$\dot{\mathbf{e}}_e = \mathbf{T}_e(\psi) (\mathbf{A}_e - \mathbf{L}) \mathbf{T}_e^T(\psi) \mathbf{e}_e + \mathbf{B}_I^e \delta_e(\hat{\nu}) - \mathbf{B} \delta_t(\nu) + \mathbf{B}_K^e \mathbf{T}_t^T(\psi) \mathbf{e}_t + \mathbf{B}_R^e [\boldsymbol{\nu}_R^T, \dot{\boldsymbol{\nu}}_R^T]^T + \mathbf{B} \mathbf{c} \quad (34)$$

where $\mathbf{e}_e \in \mathbb{R}^8 | \mathbf{e}_e := \mathbf{x} - \hat{\mathbf{x}}$ is the estimation error vector, the vectors $\delta_t(\nu)$ and $\mathbf{c}$ are defined in (23) and (8), respectively, the matrix $\mathbf{B}$ is defined in (12), the tracking error vector $\mathbf{e}_t$, the vector of references $[\boldsymbol{\nu}_R^T, \dot{\boldsymbol{\nu}}_R^T]^T$, and the transformation matrix $\mathbf{T}_t(\psi)$ are all defined in Section 4.4, and

$$\mathbf{B}_I^e := \begin{bmatrix} \mathbf{0} \\ -\mathbf{I} \end{bmatrix} \quad \text{and} \quad \mathbf{B}_R^e := \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \Delta_M \widehat{\mathbf{D}}_L & \Delta_M \widehat{\mathbf{M}} \end{bmatrix} \quad (35)$$

$$\mathbf{B}_K^e := \begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \Delta_M \mathbf{K}_I & \Delta_M \mathbf{K}_P & \Delta_M \mathbf{K}_D \end{bmatrix} \quad (36)$$

where $\Delta_M \in \mathbb{R}^{4 \times 4} | \Delta_M := \mathbf{M}^{-1} - \widehat{\mathbf{M}}^{-1}$, $\mathbf{0}, \mathbf{I} \in \mathbb{R}^{4 \times 4}$ are the null and identity matrices, and $\delta_e(\hat{\nu})$ is another mismatch vector that stems from the attempt to linearise the dynamics of (1) through state feedback under the inevitable presence of modelling errors in (14),

such that

$$\delta_e(\hat{\nu}) := \left((\mathbf{M}^{-1} \mathbf{D}_L) - (\widehat{\mathbf{M}}^{-1} \widehat{\mathbf{D}}_L) \right) \hat{\nu} \quad (37)$$

The more exactly the parameters of the CPM are known, the closer to zero the mismatch vector $\delta_e(\hat{\nu})$, the residual reference input matrix $\mathbf{B}_R^e$, and the residual control input matrix $\mathbf{B}_K^e$ are, with $(\mathbf{W}_L)_{ij} = (\mathbf{W}_M)_{ij} = 1$, $i, j \in \{1, 2, 3, 4\}$. It can be expected that the vectors $\mathbf{e}_t$, $\delta_t(\nu)$, and $[\boldsymbol{\nu}_R^T, \dot{\boldsymbol{\nu}}_R^T]^T$ remain bounded $\forall t \in [0, \infty)$, and also that they practically vanish eventually, based on the analysis and conclusion drawn in Section 4.4.

Consider the global diffeomorphism $\boldsymbol{\xi}_e := \mathbf{T}_e^T(\psi) \mathbf{e}_e$ (Khalil, 2002), based on the defined transformation $\mathbf{T}_e(\psi)$. Then, (34) can be conveniently rewritten as

$$\dot{\boldsymbol{\xi}}_e = \boldsymbol{\Xi}_e(\dot{\psi}) \boldsymbol{\xi}_e + \mathbf{B}_I^e \delta_e(\hat{\nu}) - \mathbf{B} \delta_t(\nu) + \mathbf{B}_K^e \boldsymbol{\xi}_t + \mathbf{B}_R^e [\boldsymbol{\nu}_R^T, \dot{\boldsymbol{\nu}}_R^T]^T + \mathbf{B} \mathbf{c} \quad (38)$$

where

$$\boldsymbol{\Xi}_e(\dot{\psi}) := (\mathbf{A}_e - \mathbf{L}) - \mathbf{T}_e^T(\psi) \dot{\mathbf{T}}_e(\psi) = (\mathbf{A}_e - \mathbf{L}) + \dot{\psi} \text{blockdiag}(\mathbf{T}_N, \mathbf{0}) \quad (39)$$

is a simpler LTV system matrix. The matrices $\mathbf{B}_I^e$, $\mathbf{B}$, $\mathbf{B}_K^e$, and $\mathbf{B}_R^e$ are not affected by the transformation $\mathbf{T}_e^T(\psi)$, such that the relations $\mathbf{T}_e^T(\psi) \mathbf{B}_I^e = \mathbf{B}_I^e$, $\mathbf{T}_e^T(\psi) \mathbf{B} = \mathbf{B}$, $\mathbf{T}_e^T(\psi) \mathbf{B}_K^e = \mathbf{B}_K^e$, and $\mathbf{T}_e^T(\psi) \mathbf{B}_R^e = \mathbf{B}_R^e$ hold. The relation $\mathbf{S}_e(\dot{\psi}) := \dot{\psi} \text{blockdiag}(\mathbf{T}_N, \mathbf{0}) = -\mathbf{T}_e^T(\psi) \dot{\mathbf{T}}_e(\psi)$, where $\dot{\mathbf{T}}_e(\psi) := (d/dt)[\mathbf{T}_e(\psi)]$, is based on the matrix $\mathbf{T}_N$ defined in Section 2.1. The relation $\boldsymbol{\xi}_t := \mathbf{T}_t^T(\psi) \mathbf{e}_t$ defined in Section 4.4 is also used above. The systems (34) and (38) are equivalent, although the latter shows clearer how heading angle changes affect the estimation error dynamics.

The stability of the origin $\boldsymbol{\xi}_e^* = \mathbf{0}$ (the only equilibrium point) of the unforced LTV system $\dot{\boldsymbol{\xi}}_e = \boldsymbol{\Xi}_e(\dot{\psi}) \boldsymbol{\xi}_e$ extracted from (38) is studied by suitably regarding such a system as a feedback connection between the auxiliary subsystems $\mathcal{S}_1$ and $\mathcal{S}_2$, as seen in Fig. 5, where $\mathcal{S}_1$ is an LTI dynamical system and $\mathcal{S}_2$ is an LTV memoryless function (Khalil, 2002), and then applying the circle criterion (Khalil, 2002; Lozano et al., 2000). The origin of the unforced subsystem $\mathcal{S}_1$ is GES, once the matrix $(\mathbf{A}_e - \mathbf{L})$ is Hurwitz. The matrix $(\mathbf{A}_e - \mathbf{L})$ can always be rendered Hurwitz, as the entries of $\mathbf{L}$ can be freely chosen.

Let the subsystem $\mathcal{S}_1$ be defined as indicated in Fig. 5. The matrix $\mathbf{B}_e := \mathbf{C}_e^T$, and $\mathbf{C}_e \in \mathbb{R}^{4 \times 8}$ is a sparse matrix whose only non-zero elements are $(\mathbf{C}_e)_{i,i} = 1$, $i \in \{1, 2, 3, 4\}$. Both matrices have full rank, such that $\text{rank}(\mathbf{B}_e) = \text{rank}(\mathbf{C}_e) = 4$. The matrix pair $(\mathbf{A}_e, \mathbf{B}_e)$ is controllable, since the controllability matrix $\mathbf{C}_e := [\mathbf{B}_e \ \mathbf{A}_e \mathbf{B}_e \ \mathbf{A}_e^2 \mathbf{B}_e \ \dots \ \mathbf{A}_e^7 \mathbf{B}_e]$ has full row rank, given that $\det([\mathbf{B}_e \ \mathbf{A}_e \mathbf{B}_e]) = \det(-\epsilon^{-2} \mathbf{L}_{21}) \neq 0$, for instance. The matrix pair $(\mathbf{A}_e, \mathbf{C}_e)$ is observable, as the observability matrix $\mathcal{O}_e := [\mathbf{C}_e^T \ \mathbf{A}_e^T \mathbf{C}_e^T \ (\mathbf{A}_e^2)^T \mathbf{C}_e^T \ \dots \ (\mathbf{A}_e^7)^T \mathbf{C}_e^T]^T$ has full column rank, given that $\det([\mathbf{C}_e^T \ \mathbf{A}_e^T \mathbf{C}_e^T]) = \det(\mathbf{I} - \gamma \epsilon^{-1} \mathbf{L}_{12}) \neq 0$, for instance, where $\mathbf{I} \in \mathbb{R}^{4 \times 4}$ is an identity matrix. Then, the (Hurwitz) transfer function matrix $\mathbf{H}_e(s) := \mathbf{C}_e (s\mathbf{I} - \mathbf{A}_e + \mathbf{L})^{-1} \mathbf{B}_e \in \mathbb{R}^{4 \times 4}$ associated with $\mathcal{S}_1$, where $\mathbf{I} \in \mathbb{R}^{8 \times 8}$ is an identity matrix, is SPR, as there exist for $\mathcal{S}_1$ a matrix $\mathbf{P}_e = \mathbf{P}_e^T > 0$, where $\mathbf{P}_e := \mathbf{I}$, a matrix $\mathbf{Q}_e \in \mathbb{R}^{8 \times 8} | \mathbf{Q}_e := \mathbf{I}_e \mathbf{Y}_e$, all going to be defined soon, and a constant $\epsilon_e \in \mathbb{R}_{>0} | \epsilon_e > \max(|\lambda_i(\mathbf{A}_e)|)$, $i \in \{1, 2, \dots, 8\}$, which satisfy the relations seen in (41) regarding the KYP lemma (Khalil, 2002; Lozano et al., 2000)

$$\begin{cases} (\mathbf{A}_e - \mathbf{L})^T \mathbf{P}_e + \mathbf{P}_e (\mathbf{A}_e - \mathbf{L}) = -\mathbf{Q}_e \mathbf{Q}_e^T - \epsilon_e \mathbf{P}_e \\ \mathbf{P}_e \mathbf{B}_e = \mathbf{C}_e^T \end{cases} \quad (41)$$

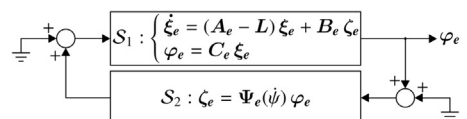


Fig. 5. Feedback connection between the auxiliary subsystems $\mathcal{S}_1$ and $\mathcal{S}_2$.

where $\Gamma_e \in \mathbb{R}^{8 \times 8}$ is an orthogonal matrix forming a suitable basis of $\mathbb{R}^8$ that consists of eigenvectors of $(A_e - L)^T + (A_e - L)$, and $\Lambda_e \in \mathbb{R}^{8 \times 8}$ is a diagonal matrix that is composed of the eigenvalues of the orthogonally diagonalisable symmetric matrix $(A_e - L)^T + (A_e - L)$ (Horn & Johnson, 2013), such that the relation $(A_e - L)^T + (A_e - L) = \Gamma_e \Lambda_e \Gamma_e^T$ holds. The diagonal matrix $\Upsilon_e \in \mathbb{R}^{8 \times 8} | \Upsilon_e > 0$, where $(\Upsilon_e)_{i,i} = \sqrt{\varepsilon_e + (\Lambda_e)_{i,i}}$, $i \in \{1, 2, \dots, 8\}$. It can also be concluded that S_1 is (state) strictly passive, once $H_e(s)$ is SPR, according to Khalil (2002) and Lozano et al. (2000).

Let the matrix $\Psi_e(\psi) \in \mathbb{R}^{4 \times 4}$ of the subsystem S_2 indicated in Fig. 5 be $\Psi_e(\psi) := \psi T_N$, where T_N is defined in Section 2.1. Then, $\varphi_e^T \xi_e = \varphi_e^T \Psi_e(\psi) \varphi_e = \xi_e^T B_e \Psi_e(\psi) C_e \xi_e = 0 \forall \psi \in \mathbb{R}$, $\forall t \in [0, \infty)$, and $\forall \varphi_e \in \mathbb{R}^4$ ($\xi_e \in \mathbb{R}^8$), due to the skew-symmetry of $\Psi_e(\psi)$. The relation $B_e = C_e^T$ is utilised above. Thus, it is immediate to conclude that $\varphi_e^T \Psi_e(\psi) \varphi_e$ lies in the sector $[0, \infty)$, and also that S_2 is a lossless passive subsystem, according to Khalil (2002) and Lozano et al. (2000).

It can thus be concluded, based on the application of the circle criterion, that ξ_e^* is GES (Th. 3.5 Lozano et al., 2000), once $\xi_e = \Xi_e(\psi) \xi_e$ is viewed as a feedback connection between the SPR transfer function matrix $H_e(s)$ (subsystem S_1) and the lossless passive subsystem S_2 , as depicted in Fig. 5.

Finally, based on the concluded global stability of ξ_e^* , thereby implying that of $e_e^* = 0$, it can be inferred that the state estimation objective $\|e_e\| = \|\mathbf{x} - \hat{\mathbf{x}}\| \rightarrow 0$ as $t \rightarrow \infty$ can always be attained. In particular, for persistent current-generated perturbations, i.e. $c \neq 0 \forall t \in [0, \infty)$, and a vanishing vector of references $[\nu_R^T, \dot{\nu}_R^T]^T$, under the inevitable presence of modelling errors, all solutions of (38), and those of (34) likewise, converge into a small ball $\mathbb{B}_e$ of radius $\rho_{\mathbb{B}_e} \in \mathbb{R}_{>0} | \rho_{\mathbb{B}_e} < 1$ ($\rho_{\mathbb{B}_e} \ll 1$ ideally) centred at $\xi_e^* = e_e^*$. The radius $\rho_{\mathbb{B}_e}$ is directly proportional to $\|c\|$ and inversely proportional to the observer gains.

5.4. Tuning

The tuning approach suggested here is heuristic and based on simulation and experimental results.

When it comes to underwater applications in general, the sensitivity to measurement errors and noise is a major concern. In particular, the horizontal position measurements, which are typically provided by hydroacoustic positioning systems, have unfavourable characteristics, e.g. low update rates (< 1 Hz), and jumpy and frozen measurements. The linear velocity measurements, when available, are typically provided by Doppler velocity logs (DVLs) under low update rates (≈ 1 Hz). On the other hand, the controlled plants typically have (very) slow and overdamped dynamics. Therefore, an HGSO can be satisfactorily tuned with relatively high gains.

The output injection matrix L in (33) damps the estimation response in such a manner that a certain trade-off between relying more upon the response that is predicted based on the CPM, or relying more upon the incoming measurements instead, can be exploited. Indeed, such kind of trade-off is a common feature of all state observation algorithms. It is suggested to tune

$$L_{ij} := \kappa_{ij} \hat{M}^{-1} \quad (42)$$

where the constants $\kappa_{11}, \kappa_{12}, \kappa_{21}, \kappa_{22} \in \mathbb{R}_{>0} | \kappa_{21} \geq \kappa_{11} \in [1, 5]$ and $\kappa_{22} \geq \kappa_{12} \in [1, 50]$. In addition, $\kappa_{12} \geq \kappa_{21}$, when velocity measurements are used by the HGSO, i.e. when $\gamma = 1$.

5.5. Stability of the output feedback control system

The stability of the output feedback control system is studied based on the previous results. The expressions of the tracking error dynamics in (24) and of the estimation error dynamics in (38) are

of particular interest here. The expressions (19) and (34) could have been optionally chosen for such task instead. The output feedback control system can be expressed as

$$\dot{\xi}_c = \Xi_c(\psi) \xi_c + B_c^T [(\delta_t(\nu) - c)^T, \delta_e^T(\dot{\nu})^T]^T + B_c^T [\nu_R^T, \dot{\nu}_R^T]^T \quad (43)$$

where $\xi_c := [\xi_t^T, \xi_e^T]^T \in \mathbb{R}^{20}$ is the state vector, $[(\delta_t(\nu) - c)^T, \delta_e^T(\dot{\nu})^T]^T \in \mathbb{R}^8$ is the vector of perturbations (internal and external), $[\nu_R^T, \dot{\nu}_R^T]^T$ is the same vector of references used before, and the matrices are defined as

$$\Xi_c(\psi) := A_c + S_c(\psi) \quad \text{and} \quad A_c := \begin{bmatrix} A_t & \mathbf{0}_{12 \times 8} \\ B_K^e & A_e - L \end{bmatrix} \quad (44)$$

$$S_c(\psi) := \psi \text{blockdiag}(T_N, T_N, \mathbf{0}, T_N, \mathbf{0}) \quad (45)$$

$$B_c^T := \begin{bmatrix} B_t^T & \mathbf{0}_{12 \times 4} \\ -B & B_f^T \end{bmatrix} \quad \text{and} \quad B_c^T := \begin{bmatrix} B_R^T \\ B_K^T \end{bmatrix} \quad (46)$$

where $\mathbf{0} \in \mathbb{R}^{4 \times 4}$ and $\mathbf{0}_{12 \times j} \in \mathbb{R}^{12 \times j}$, $j \in \{4, 8\}$, are null matrices. The other matrices are already defined either in Section 4.4 or in Section 5.3. The matrix B_K^e would ideally be null, implicating that A_c would ideally become a block diagonal matrix. Notice that the separation principle (Franklin et al., 2009; Khalil, 2002; Friedland, 1986) is manifested in (43) and (44) through the cascade structure of the matrix A_c in (44).

The stability of the origin $\xi_c^* = \mathbf{0}$ (the only equilibrium point) of the unforced LTV system $\dot{\xi}_c = \Xi_c(\psi) \xi_c$ extracted from (43) is studied by suitably regarding such a system as a feedback connection between the auxiliary subsystems S_1 and S_2 , as seen in Fig. 6, where S_1 is an LTI dynamical system and S_2 is an LTV memoryless function (Khalil, 2002), and then applying the circle criterion (Khalil, 2002; Lozano et al., 2000). The origin of the unforced subsystem S_1 is GES, once the lower triangular matrix A_c in (44) is Hurwitz. The matrix A_c can always be rendered Hurwitz, since the matrices A_t and $(A_e - L)$ can always be individually rendered Hurwitz likewise, as concluded in Sections 4.4 and 5.3, respectively.

Let the subsystem S_1 be defined as indicated in Fig. 6. The matrix $B_c := C_c^T$, and $C_c \in \mathbb{R}^{8 \times 20}$ is a sparse matrix whose only non-zero elements are $(C_c)_{1,1} = (C_c)_{2,2} = (C_c)_{3,5} = (C_c)_{4,6} = (C_c)_{5,13} = (C_c)_{6,14} = (C_c)_{7,15} = (C_c)_{8,16} = 1$. Both matrices have full rank, such that $\text{rank}(B_c) = \text{rank}(C_c) = 8$. The matrix pair (A_c, B_c) is controllable, since the controllability matrix $C_c := [B_c \ A_c B_c \ A_c^2 B_c \ \dots \ A_c^{19} B_c]$ has full row rank, provided that $\det([B_c \ A_c^2 B_c \ A_c^4 B_c]) \neq 0$, for instance. The matrix pair (A_c, C_c) is observable, since the observability matrix $O_c := [C_c^T \ A_c^T C_c^T \ (A_c^2)^T C_c^T \ \dots \ (A_c^{19})^T C_c^T]^T$ has full column rank, given that $\det([C_c^T \ (A_c^2)^T C_c^T \ (A_c^4)^T C_c^T]) \neq 0$, for instance. Recall that the hydrodynamic derivatives in D_t and M_A , therefore those in M , are all expected to be non-zero and distinct, as stated in Section 2.1. This fact guarantees that C_c and O_c have full rank, because A_c is composed of A_t , which in turn is composed of D_t and M^{-1} . Then, the (Hurwitz) transfer function matrix $H_c(s) := C_c(sI - A_c)^{-1} B_c \in \mathbb{R}^{8 \times 8}$ associated with S_1 , where $I \in \mathbb{R}^{20 \times 20}$ is an identity matrix, is SPR, as there exist for S_1 a matrix $P_c = P_c^T > 0$, where $P_c := I$, a matrix $Q_c \in \mathbb{R}^{20 \times 20} | Q_c := \Gamma_c \Upsilon_c$, all going to be defined soon, and a constant $\varepsilon_c \in \mathbb{R}_{>0} | \varepsilon_c > \max(|\lambda_i(\Lambda_c)|)$, $i \in \{1, 2, \dots, 20\}$, which satisfy the relations seen in (47) regarding the KYP lemma (Khalil, 2002; Lozano et al., 2000)

$$\begin{cases} A_c^T P_c + P_c A_c = -Q_c Q_c^T - \varepsilon_c P_c \\ P_c B_c = C_c^T \end{cases} \quad (47)$$

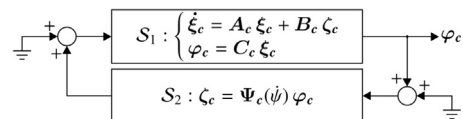


Fig. 6. Feedback connection between the auxiliary subsystems S_1 and S_2 .

where $\Gamma_c \in \mathbb{R}^{20 \times 20}$ is an orthogonal matrix forming a basis of $\mathbb{R}^{20}$ that consists of eigenvectors of $(A_c^T + A_c)$, and $\Lambda_c \in \mathbb{R}^{20 \times 20}$ is a diagonal matrix that is comprised of the eigenvalues of the orthogonally diagonalisable symmetric matrix $(A_c^T + A_c)$ (Horn & Johnson, 2013), such that $A_c^T + A_c = \Gamma_c \Lambda_c \Gamma_c^T$. The diagonal matrix $\Upsilon_c \in \mathbb{R}^{20 \times 20} | \Upsilon_c > 0$, where $(\Upsilon_c)_{i,i} = \sqrt{\varepsilon_c + (\Lambda_c)_{i,i}}$, $i \in \{1, 2, \dots, 20\}$. It can also be concluded that S_1 is (state) strictly passive, once $H_c(s)$ is SPR, according to Khalil (2002) and Lozano et al. (2000).

Let the matrix $\Psi_c(\psi) \in \mathbb{R}^{8 \times 8}$ of the subsystem S_2 indicated in Fig. 6 be defined as $\Psi_c(\psi) = \text{blockdiag}(\Psi_t(\psi), \Psi_e(\psi))$, where the matrices $\Psi_t(\psi)$ and $\Psi_e(\psi)$ are defined within Sections 4.4 and 5.3, respectively. Then, $\varphi_c^T \xi_c = \varphi_c^T \Psi_c(\psi) \varphi_c = \xi_c^T B_c \Psi_c(\psi) C_c \xi_c = 0 \forall \psi \in \mathbb{R}$, $\forall t \in [0, \infty)$, and $\forall \varphi_c \in \mathbb{R}^8$ ($\xi_c \in \mathbb{R}^{20}$), due to the skew-symmetry of $\Psi_c(\psi)$. The relation $B_c = C_c^T$ is utilised above. Thus, it is immediate to conclude that $\varphi_c^T \Psi_c(\psi) \varphi_c$ lies in the sector $[0, \infty)$, and also that S_2 is a lossless passive subsystem, according to Khalil (2002) and Lozano et al. (2000).

It can thus be concluded, based on the application of the circle criterion, that ξ_c^* is GES (Th. 3.5 Lozano et al., 2000), once $\xi_c = \Xi_c(\psi) \xi_c$ is viewed as consisting of a feedback connection between the SPR transfer function matrix $H_c(s)$ (subsystem S_1) and the lossless passive subsystem S_2 , as depicted in Fig. 6. The global exponential stability concluded for $\xi_c^* = 0$ is the main result sought and satisfactorily found here.

For persistent current-generated perturbations, i.e. $c \neq 0 \forall t \in [0, \infty)$, and a vanishing vector of references $[\nu_R^T, \dot{\nu}_R^T]^T$, under the inevitable presence of modelling errors, all solutions of (43) converge into a small ball $\mathbb{B}_c$ of radius $\rho_{\mathbb{B}_c} \in \mathbb{R} > 0 | \rho_{\mathbb{B}_c} < 1$ ($\rho_{\mathbb{B}_c} \ll 1$ ideally) centred at ξ_c^* . The radius $\rho_{\mathbb{B}_c}$ is directly proportional to $\|c\|$ and inversely proportional to the gains of the MIMO PID controller and of the HGSO.

6. Results from full-scale sea trials

This section presents selected results from full-scale sea trials in which the NTNU's ROV Minerva was navigated in Trondheimsfjorden, Norway, in April 2014 by the proposed MCS. Trajectory tracking and dynamic positioning tests were carried out. The references used were generated as described in Section 3. Information regarding the test setup and the ROV Minerva can be found in Appendix A. Details on the implementation of the MCS in practice are found in Fernandes et al. (2014).

6.1. Trajectory tracking test

Figs. 7–10 depict results from a test in which the spatial reference trajectory consisted of the so-called lawn-mower pattern describing a planar path in the horizontal plane elevated a constant height from the sea bottom, i.e. it replicated the sea bottom profile. The lawn-mower pattern was composed of three 20 m-long parallel legs separated 2 m from each other, see Fig. 7. All measurements were reasonably well filtered by the HGSO, thereby yielding the acceptable tracking performance seen in Figs. 7–9 and the smooth control action seen in Fig. 10. Table 1 summarises the information contained in Figs. 7 and 8, where ‘pos.’ and ‘h.a.’ stand for position and heading angle. The table shows position error norms and absolute heading angle errors calculated with respect to the corresponding references. The largest errors occurred during the turns, i.e. heading angle changes, due to the relatively high maximum angular velocity reference used. Fig. 10 shows that the starboard and port thrusters nearly reached their saturation limits during the turns. A jump in the horizontal position measurement occurred during the last turn around 196.6 s. Notice the consistently better tracking performance attained along the first and third legs than that achieved along the second leg. The trajectory tracking errors remained close to the stated

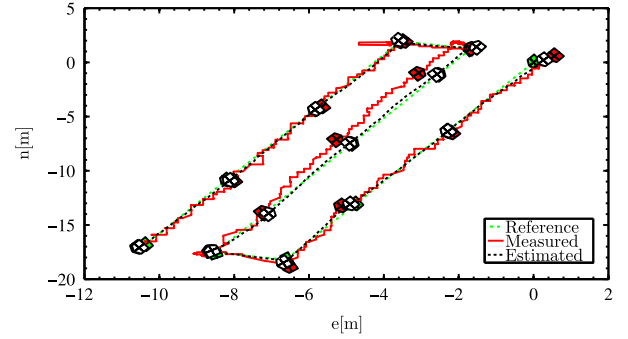


Fig. 7. Trajectory tracking test: top view of the NE-plane.

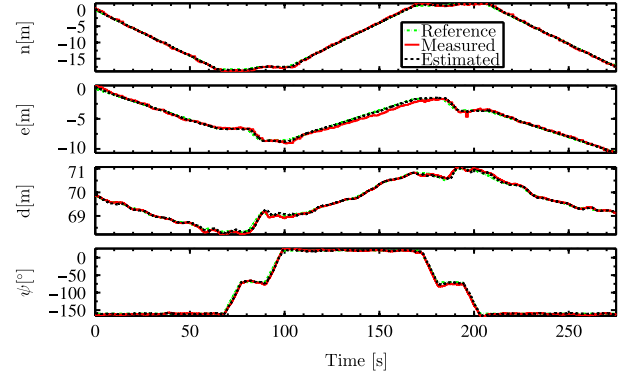


Fig. 8. Trajectory tracking test: position and heading angle (NED frame).

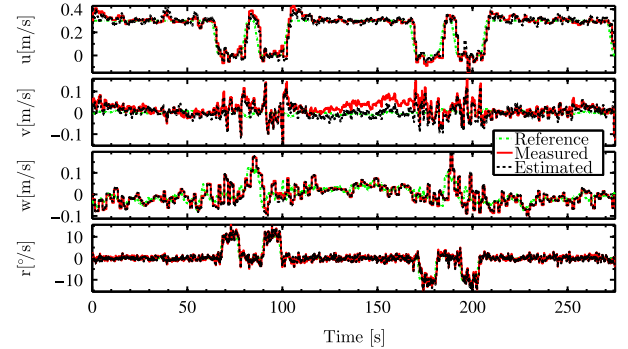


Fig. 9. Trajectory tracking test: velocities (BF frame).

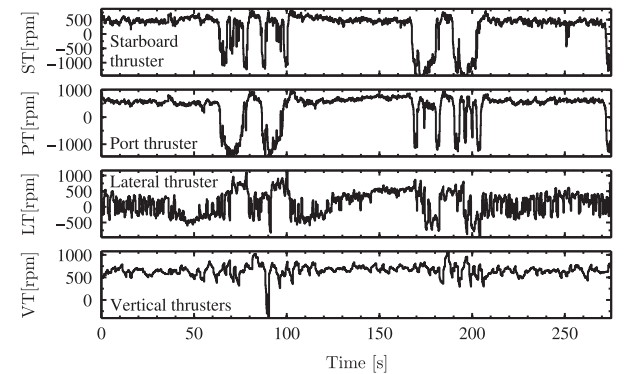


Fig. 10. Trajectory tracking test: commanded propeller rotations.

mean values during the motion along the first and third legs, whereas a pronounced error occurred during the motion along the second leg. A simultaneous sway velocity deviation can be seen in Fig. 9. Such tracking error was most probably caused by tuning issues regarding

Table 1
Information summary concerning Figs. 7 and 8.

	Estimation		Measurement	
	Max error	Mean error	Max error	Mean error
pos.	49.3 cm (@ 186.3 s)	15.9 cm	1.168 m (@ 196.6 s)	37.6 cm
h.a.	9.49° (@ 93.9 s)	1.74°	12.85° (@ 181.6 s)	2.41°

both the controller and the observer. Firstly, the output injection matrix $\mathbf{L}$ in (33) overdamped the estimation, which relied more on the response predicted based on the CPM than on the incoming measurements. Secondly, the combination of the small tracking errors calculated based on the overdamped state estimation and the low controller gains precluded the possibility of achieving better tracking performance. Other factors, e.g construction asymmetries of Minerva, modelling simplifications, plant parameter variations, and disturbances due to the sea current, also played important roles in the overall trajectory tracking performance.

The controller was tuned as follows:

$$\begin{cases} \mathbf{W}_\Omega = \text{diag}(690, 495, 330, 60) \\ (\mathbf{W}_C)_{ij} = (\mathbf{W}_L)_{ij} = (\mathbf{W}_M)_{ij} = (\mathbf{W}_Q)_{ij} = 1 \\ \mathbf{W}_P = \mathbf{I} \end{cases} \quad (48)$$

where $\mathbf{I} \in \mathbb{R}^{5 \times 5}$ is an identity matrix, and $i, j \in \{1, 2, 3, 4\}$.

The HGSO was tuned as follows:

$$\begin{cases} \kappa_{11} = 2, \quad \kappa_{12} = 20, \quad \kappa_{21} = 4, \quad \text{and} \quad \kappa_{22} = 40 \\ \epsilon = 0.001 \quad \text{and} \quad \gamma = 1 \end{cases} \quad (49)$$

6.2. Dynamic positioning test

Figs. 11–13 depict results from a DP test in which the ROV was desired to keep station at a depth of 70 m. Acceptable state regulation can be seen in Figs. 11 and 12 and reasonably smooth control action can be seen in Fig. 13. Table 2 summarises the information contained in Fig. 11. The errors seen in Table 2 are larger than those seen in Table 1 only because of the jump in the horizontal position measurement that occurred around 319 s. Another jump occurred earlier, around 130 s. The errors remained close to the stated mean values except when measurement jumps occurred. Notice that there is no detectable increase in the north position tracking error, despite the non-zero mean sway velocity indicated in Fig. 12. In particular, it is easy to realise that the small mean sway velocity would have resulted in a mean lateral force through the derivative gain of the MIMO PID controller, and also that such force would have caused the ROV to drift sideways. Nevertheless, such (constant) disturbing force was successfully counteracted by the integral term of the MIMO PID controller, as expected. This conflict led to some undesirable and unnecessary extra propeller work, as it can be seen in Fig. 13, which explains the existence of the unsteady lateral motion of Minerva, which is manifested through the (high-variance) sway velocity in consonance with the unsteady (zero-mean) displacement around the origin of the N-axis seen in Fig. 11. Notice that, in particular, the lateral thruster, see Fig. A1, worked relatively more than the two other horizontal thrusters. The most probable causes of such behaviour were the same as in the trajectory tracking test. On the other hand, the HGSO captured more accurately the plant dynamics in this case, as already expected, since $\nu \approx \mathbf{0}$ during all the test.

Table 2
Information summary concerning Fig. 11.

	Estimation		Measurement	
	Max error	Mean error	Max error	Mean error
pos.	60.2 cm (@ 319.2 s)	9.6 cm	1.368 m (@ 318.2 s)	14.8 cm
h.a.	3.88° (@ 319.8 s)	0.86°	3.83° (@ 319.6 s)	0.88°

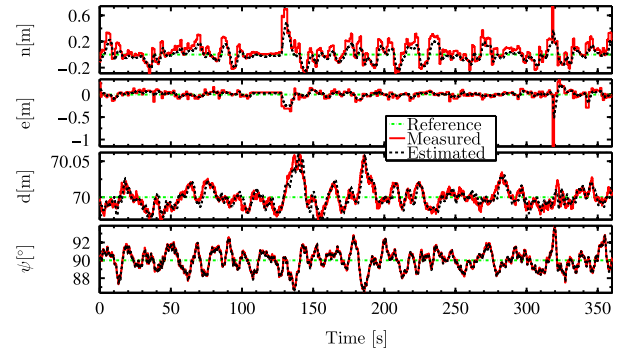


Fig. 11. DP test: position and heading angle (NED frame).

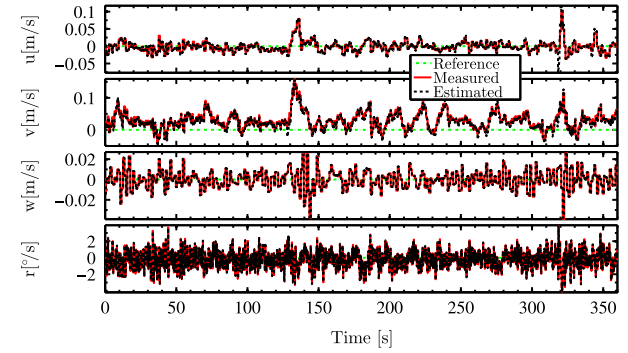


Fig. 12. DP test: velocities (BF frame).

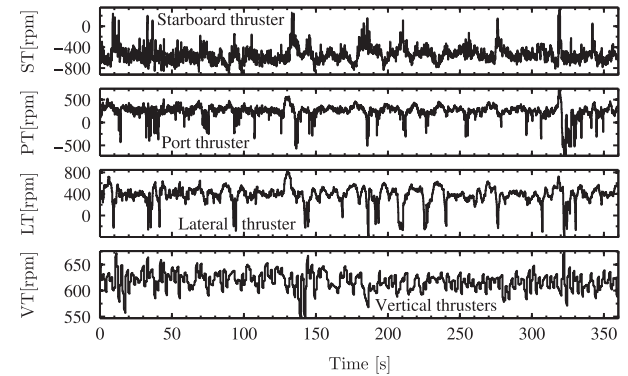


Fig. 13. DP test: commanded propeller rotations.

The controller was tuned as follows:

$$\begin{cases} \mathbf{W}_\Omega = \text{diag}(1150, 830, 555, 100) \\ (\mathbf{W}_C)_{ij} = (\mathbf{W}_L)_{ij} = (\mathbf{W}_M)_{ij} = (\mathbf{W}_Q)_{ij} = 1 \\ \mathbf{W}_P = \mathbf{I} \end{cases} \quad (50)$$

where $\mathbf{I} \in \mathbb{R}^{5 \times 5}$ is an identity matrix, and $i, j \in \{1, 2, 3, 4\}$.

The HGSO was tuned as follows:

$$\begin{cases} \kappa_{11} = 2, & \kappa_{12} = 20, & \kappa_{21} = 3, & \text{and} & \kappa_{22} = 50 \\ \epsilon = 0.002 & \text{and} & \gamma = 1 \end{cases} \quad (51)$$

7. Concluding remarks

An MCS for observation class ROVs was developed, tested and analysed in the paper. Conducted full-scale sea trials confirmed that the MCS is able to perform stably and satisfactorily under the challenging operating conditions faced in practice, e.g. unmodelled plant dynamics, plant parameter variations, measurement errors and noise, and environmental disturbances.

In particular, the experimental results encourage the use of an HGSO as another replacement alternative to the benchmark EKF in MCSs for observation class ROVs.

Based on the analysis of simulation and experimental results, it is possible to conclude that improved motion control performance can be achieved under more favourable operating conditions, namely (i) higher update rates when it comes to the horizontal position measurements; (ii) more accurately identified CPM parameters; and (iii) proper thruster modelling and control. Clearly, higher sensor update rates yield better performance, as it can be verified in the experimental results regarding the heave and yaw motion. This evidence is corroborated by simulation results. While the update rates of the hydroacoustic positioning systems are currently low, there may be solutions in the pipeline, e.g. obtaining faster and more accurate position measurements through the geometry of the umbilical cable, e.g. Hiranandani, White, Clark, Gambin, and Buhagiar (2009). Accurately identified model parameters doubtlessly have a high, positive impact on the performance of model-based MCSs. The experimental results were influenced by the lower accuracy of the identified model parameters that occurs because the ROV Minerva serves different purposes when it comes to the different research requirements and needs of the researchers at NTNU. Moreover, further insight about how to fine-tune the HGSO could have been gained through the availability of a more detailed and accurately identified CPM. Poor thruster modelling and control is certainly an important hindrance to better motion control performance. While it is very difficult to model underwater thrusters accurately in practice, it seems reasonable to expect enhanced performance if the thruster servos of the ROV Minerva were able to deliver commanded torques rather than the commanded propeller shaft angular velocities. Propeller torque control is surely a goal to be attained for better motion control performance.

Future works must seek to find more effective tuning methods for the HGSO, and means of accounting for the effects of unmodelled plant dynamics, plant parameter variations, and environmental disturbances, e.g. to include bias estimators, in order to avoid filter divergences.

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Appendix A. NTNU's ROV Minerva

Minerva is a SUB-fighter 7500 ROV manufactured by Sperre AS for NTNU in 2003. The NTNU's research vessel (R/V) Gunnerus

(<http://www.ntnu.edu/marine/gunnerus>) is the support vessel used to operate Minerva. The ROV is powered from, and communicates with, R/V Gunnerus through a 600 m-long umbilical cable. It has five thrusters with fixed pitch propellers. The starboard and port thrusters are oriented 10° towards the longitudinal axis. The lateral thruster is the only that has one propeller at each end of its shaft, whereas the others have a single propeller each. Fig. A1 shows the thruster installation configuration. Table A1 keeps on the specifications. Details are found in Sørensen et al. (2012) and Dukan, Ludvigsen, and Sørensen (2011).

The (four-DoF CPM) parameters of Minerva are

$$\begin{cases} \bar{\mathbf{M}}_{RB} = \text{diag}(460, 460, 460, 50) \\ \bar{\mathbf{M}}_A = \text{diag}(293, 302, 326, 57) \\ \bar{\mathbf{D}}_L = \text{diag}(29, 41, 64, 25) \\ \bar{\mathbf{D}}_Q = \text{diag}(292, 584, 635, 100) \end{cases} \quad (52)$$

A high-precision hydroacoustic positioning system model HiPAP 500 by Kongsberg Maritime AS determines the north n and east e position components of Minerva relative to R/V Gunnerus with accuracy better than 0.1 m at update rates ≤ 1 Hz. The depth d (down position component), see Fig. 2, is determined based on the measurements provided by the precision, temperature compensated, piezo-resistive underwater pressure sensor model miniIPS 0760001-100 by Valeport Ltd. It has full-scale span 100 bar, accuracy ± 10 mbar, and resolution 1 mbar. Its maximum output update rate is 8 Hz. The heading angle ψ and the yaw rate r are determined through the use of a dedicated complementary filter (Mahony, Hamel, & Pflimlin, 2008) that treats the measurements provided by the micro-electro-mechanical system (MEMS)-based inertial measurement unit (IMU) model MTi-100 by Xsens Technologies B.V. The gyroscopes of the IMU have typical full-scale spans $450^\circ/\text{s}$, maximum bias repeatability $0.2^\circ/\text{s}$ (1 year), and typical in-run bias stability $10^\circ/\text{h}$. Its accelerometers have typical full-scale spans 50 m/s^2 , maximum bias repeatability 30 mm/s^2 (1 year), and typical in-run bias stability $40 \mu\text{g}$. The maximum output update rate is 2 kHz with latency < 2 ms. The velocities u ,

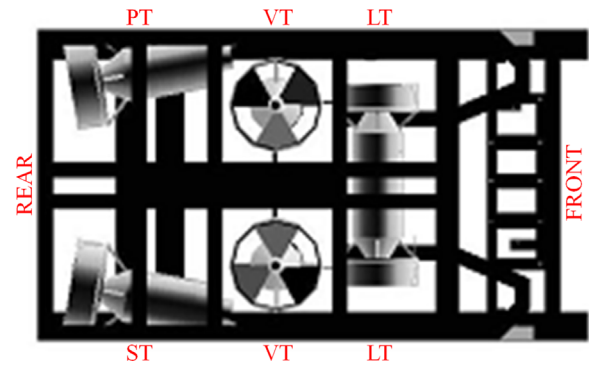


Fig. A1. Top view of the thruster installation configuration of Minerva (Legend: L, lateral; P, port; S, starboard; T, thruster; V, vertical).

Table A1
Basic specifications of Minerva.

Dimensions	144 × 82 × 81 [cm] (L × W × H)
Weight (air)	485 kg
Max payload	20 kg
Max depth	700 m
Thrusters	Horizontally installed: 3 × 2 kW Vertically installed: 2 × 2 kW
Thrust capacity (min/max)	Surge: – 220 N/480 N Sway: – 195 N/195 N Heave: – 180 N/390 N Yaw: – 300 Nm/300 Nm

v , and w are simultaneously measured by a 1200 kHz Workhorse Navigator DVL by Teledyne RD Instruments, Inc. It has full-scale spans 10 m/s, long-term accuracy $\pm 0.2\% \pm 1$ mm/s, and resolution 1 mm/s. Its maximum output update rate is ≤ 1 Hz.

The MCS is implemented on a cRIO, and programmed via LabVIEW 2012 – SP1 for Microsoft Windows, both by National Instruments. The MCS runs at the constant sampling frequency of 6.6 Hz (sampling period of 150 ms). Further details can be found in Sørensen et al. (2012) and Dukan et al. (2011).

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